

Unveiling coherent structures in turbulent molecular clouds

S. Delcamp^{1,2}, A. Perroni¹, and P. Hily-Blant¹

¹ Univ. Grenoble Alpes, CNRS, IPAG, 38000 Grenoble, France e-mail: simon.delcamp@univ-grenoble-alpes.fr, pierre.hily-blant@univ-grenoble-alpes.fr

² Laboratoire de Physique de l'Ecole Normale Supérieure, ENS, Université PSL, CNRS, Sorbonne Université, Université de Paris, Paris, France

ABSTRACT

Turbulence in molecular clouds plays a key role in regulating the star formation rate which is currently, in the star formation history of the universe, at a still unexplained low value. However, being strongly supersonic and magnetized, and in competition with gravity, the large body of knowledge available from terrestrial turbulence does not transpose readily to the framework of gravo-turbulence. Characterizing the properties of molecular clouds turbulence is thus of paramount importance. In this context, the tools developed for the analysis of incompressible turbulence need to be adapted to the specific physical conditions of molecular clouds and to the techniques used to observe them. In this work, we present a new tool, `Unveil.jl`, dedicated to the analysis of the turbulent velocity field of molecular clouds measured through radio astronomical heterodyne position-position-velocity cubes. **The primary outcome is the use of two independent methods for the noise reduction of the data, allowing a precise analysis of the velocity field. One of the two methods is based on Principal Component Analysis, which is used as a data compression tool, while the second method extract the signal part from the noise of the spectra. These cleaning conduct to the computation of more precise centroid velocity for each spectrum of at least 20%, from which an ensemble of statistical analysis is performed, including velocity gradients and velocity structure function.** Another central and original feature of the `Unveil.jl` code is the convergence study of the velocity structure function calculations. Finally, velocity structure function exponents and their uncertainties are measured, allowing the molecular clouds turbulent cascade to be characterized. Each step of the data processing in `Unveil.jl` has been tested and characterized using both real and modeled CO(1-0) observations of molecular clouds, based on realistic numerical simulations. Furthermore, various aspects developed in `Unveil.jl` find broader applications, especially in the context of the data reduction of large heterodyne hyperspectral cubes, such as robust automatic windowing of spectra.

1. Introduction

The star formation rate in the Milky Way and other galaxies is currently at a low value (Krumholz 2014; Madau & Dickinson 2014), and the reasons for this are still elusive. More specifically, the timescale for converting the molecular gas into stars is two orders of magnitude longer than the free-fall time. This could result from two extreme scenarios: one in which the conversion of molecular gas into stars is intrinsically slower than what models can predict, because we are missing (or neglecting) some processes; in the other, the conversion is fast but intrinsically inefficient, and only a fraction of the molecular gas turns into stars (Girichidis et al. 2020). The reality may also lie between these two views.

Although the main processes involved in the evolution of the molecular gas have been brought to light (gravity, magnetic fields, turbulence, and stellar feedback), star formation rates from numerical simulations are still factors of ≈ 4 too large (Federrath 2015; Chevance et al. 2020). Indeed, many details are still poorly known, such as the dominant turbulence dissipation processes (Richard et al. 2022) or the intensity and morphology of the magnetic fields in diffuse or dense regions of molecular clouds (Falgarone et al. 2008; Crutcher 2012). Furthermore, the general properties of molecular clouds turbulence are also elusive. Indeed, most theoretical results in turbulence hardly cover the parameter space of molecular clouds where turbulent motions are quite extreme compared to terrestrial (natural, laboratory) or numerical experiments: strongly supersonic, with sonic Mach number, $\mathcal{M}_S$, as high as ≈ 10 , Reynolds numbers $Re \sim 10^7$, Prandtl number $\nu/\eta \sim 10^4$, magnetized fluid ranging

from the sub- to super-Alfvénic regimes depending on scales, bistable equation of state. Thus, building on the existing body of knowledge to characterize the properties of molecular cloud turbulence still requires some extrapolation beyond the charted territory.

Fluid velocity is the most fundamental quantity in any turbulence theory. In molecular clouds, this quantity is only partially, and indirectly, available. Yet, it is measured with extremely high accuracy **changed the "v" by ν** ($\delta\nu/\nu \sim 10^{-9}$) and precision, and has enabled significant achievements. Thus, a relation between the velocity dispersion and length scale, showing a power-law over one decade, has been observed in molecular clouds and giant molecular clouds (Falgarone et al. 2009; Heyer & Dame 2015), although with a large scatter. This power-law is usually interpreted in the framework of the theory of turbulence first proposed by Kolmogorov (1941) (K41), although the exponent does not match. A more fundamental property, namely the intermittency of the turbulent cascade in the inertial range (Anselmetti et al. 2001), has also been demonstrated in molecular clouds through its signature on the statistics of the velocity field (Lis et al. 1996; Pety & Falgarone 2003; Hily-Blant et al. 2008; Hily-Blant & Falgarone 2009). Nevertheless, despite their chaotic nature, turbulent motions preserve a degree of spatial coherence, and structures, such as filaments or clusters of filaments, have been observed (Douady et al. 1991; Moisy & Jiménez 2004). Although similar behaviors have been reported in the Polaris Molecular Cloud (Hily-Blant et al. 2008), little is known about coherent structures of turbulence in molecular clouds, a field that may be considered in its infancy. More generally, the coherent structures of turbulence are created by the velocity field, and the

statistical and structural aspects of turbulence are two facets of one single phenomenon, at the heart of which lies the turbulent velocity field. Measuring this quantity is therefore of paramount importance to characterize molecular clouds turbulence.

Turbulence is best understood through statistical analysis. However, even in controlled experiments, measuring the statistical properties of the velocity field comes with challenges. A salient difficulty is the requirement of data sets sufficiently large to allow high-order moments—which focus on extreme, hence rare, events—to be computed. The criteria for the convergence of such moments are still an open question although empirical methods exist (Belin et al. 1996; Dudok De Wit 2004; DeMarco & Basu 2017). The limited size of the sample also affects our ability to identify the inertial range of turbulence where scaling laws are expected to hold. Indeed, the extent of the inertial domain increases with the Reynolds number, and very high values are difficult to obtain, in controlled experiments or numerical simulations (Elsinga et al. 2020). An empirical method, known as Extended-Self-Similarity (ESS) (Benzi et al. 1993), alleviates considerably the issue, although for reasons that remain theoretically unclear (Arneodo et al. 1996; Chakraborty et al. 2010; Kiliyanpilakkil & Basu 2015).

Characterizing the properties of molecular clouds is even more challenging than in experiments. Nevertheless, the ESS allowed Hily-Blant et al. (2008) to peer into the properties of the turbulent cascade, by measuring the high-order statistics of the velocity field in the Polaris molecular cloud which supports an incompressible-like turbulence dominated by solenoidal motions (Federrath et al. 2010). Yet, most studies of the velocity field are limited to the second-order moment (its power spectrum) although most details of the turbulent cascade are contained into the higher orders. In the context of measuring the statistical properties of the velocity field, a crucial aspect is the uncertainties on the derived measurements and properties. Another central issue is the convergence of the high-order moments of the velocity field. Finally, in their analysis of the two-point statistics of the velocity field, Hily-Blant et al. (2008) noticed that the largest variations velocity gradients, which are associated with the intermittent regions of turbulence dissipation, are associated with variations of the ^{12}CO line wing emission, raising the need for a data reduction and analysis methods that best preserve these wings. These questions are however poorly explored and constitute the chief objective of the present work.

The adopted philosophy is to develop two independent methods to measure the velocity field and its statistical properties, from which uncertainties can be estimated. We also provide a new mean to check the convergence of high-order moments. The overall methodology also is original, since the validation of the new methods rests upon the use of realistic simulations of molecular clouds observations, taking benefits of public databases of (HD and MHD) 3D numerical simulations (Burkhart et al. 2020). The resulting tools open new possibilities to enlarge our knowledge on the turbulence in molecular clouds, hence on the universal properties of star formation.

The paper is organized as follows. Section 2 provides the definitions of the fundamental quantities involved in the statistical analysis of the turbulent velocity field. Sections 4 and 5 explain the two centroid velocity calculation methods. The Section 6 briefly motivates the choice of the Julia language. In Section 7, we present the code benchmark on simulated hyperspectral cubes of molecular clouds. Then, Section 8 is dedicated to the application of `Unveil.jl` on the Polaris data, with a comparison with earlier results. Specific aspects of the code, and more general topics on the statistical analysis of interstellar turbulent

velocity fields are developed in Section 9. Section 10 summarizes the main results and provide some **concluding** statements.

2. General considerations and definitions

Radio astronomical observations provide hyperspectral maps with, at each position (x, y) on the plane of sky, a spectrum giving the spectral line intensity $T(x, y, v)$ as a function of the Doppler shift velocity v . These maps are often referred to as position-position-velocity (PPV) cubes. The turbulent velocity field, which is the primary quantity in turbulence analysis, is not directly accessible in molecular clouds. Instead, the primary observable in PPV cubes are spectra which result from the non-linear and non-local convolution of the density and velocity fields, and some mapping has to be applied to the spectra in order to extract a meaningful velocity measurement (Koch et al. 2017). However, it has been shown that the velocity centroid (CV), defined as the intensity-weighted first order moment, provides a good proxy of the velocity field (Lis et al. 1996; Pety 1999; Pety & Falgarone 2000; Pety & Falgarone 2003). Yet, any spectrum is the result of a complex convolution of the density and velocity fields, so that there is no simple mapping between a measured CV and the line-of-sight velocity component of the gas $u_z(x, y, z)$. In regularly sampled spectra, the CV at position x, y in the map reads:

$$\text{CV}(x, y) [\text{km/s}] = \frac{1}{W(x, y)} \sum_{i=1}^N T(x, y, v_i) v_i \quad (1)$$

where $W(x, y) = \sum_{i=1}^N T(x, y, v_i)$ is the integrated intensity and N is the number of spectral channels. From a PPV cube one thus obtains a two-dimensional CV map.

Theoretical studies of turbulent flows heavily rely on the two-point statistics of the turbulent velocity field through the velocity increments $\delta \mathbf{u}(\mathbf{r}, \ell) = \mathbf{u}(\mathbf{r} + \ell) - \mathbf{u}(\mathbf{r})$ (longitudinal increments have $\ell \parallel \mathbf{u}$). The equivalent of $\delta \mathbf{u}$ in the observational domain of molecular clouds will be a difference of CV called centroid velocity increment (CVI). The CVI over a lag ℓ is

$$\Delta \text{CV}_\ell(\mathbf{r}) [\text{km/s}] = \text{CV}(\mathbf{r} + \ell) - \text{CV}(\mathbf{r}), \quad \mathbf{r} = (x, y), \quad (2)$$

A CVI is a difference, restricted to the plane of sky, of a proxy of u_z , and is thus closer to the transverse ($\ell \perp \mathbf{u}$), rather than longitudinal, velocity increment.

Structure functions of orders p are often used to characterize the turbulent cascade; they are defined as $S_p(\ell) = \langle |\delta \mathbf{u}(\mathbf{r}, \ell)|^p \rangle_{\mathbf{r}, \ell}$ where the average is done over all positions $\mathbf{r}$ and all orientations of ℓ . The averaging may be performed either directly or using the probability density function (PDF) of the velocity increments. The **PDF** approach mitigates the impact of any outlier, especially at high orders p , by offering the possibility to filter out bins that consist in a number of events smaller than some threshold. In practice, in this work, structure functions are computed using the absolute value of the CVI (Hily-Blant et al. 2008):

$$S_p(\ell) = \langle \Delta \text{CV}_\ell \rangle_{\mathbf{r}} = \int_{-\infty}^{\infty} \mathcal{P}(\Delta \text{CV}_\ell) |\Delta \text{CV}_\ell|^p d\Delta \text{CV}_\ell, \quad (3)$$

with $\mathcal{P}(\Delta \text{CV}_\ell)$ the PDF of the CVI, with the usual normalization condition $\int \mathcal{P}(x) dx = 1$.

Structure functions are a fundamental tool for turbulence studies and can be used to different purposes. One of the key results of the K41 theory (Kolmogorov 1941) is that structure functions are power-laws of the lag, $S_p(l) \propto l^{p/3}$, with $\zeta_p = p/3$

for homogeneous turbulence. However, the cascade is known to be intermittent, a universal property that applies to all regimes of turbulence, including supersonic (Porter et al. 2002; Kritsuk et al. 2007) and molecular clouds (Lis et al. 1996; Pety & Falgarone 2003; Hily-Blant & Falgarone 2009). One salient signature of intermittency are the tails of the PDF of velocity increments at small lags, that become increasingly strong as ℓ decreases. This reflects the higher occurrence of extreme increments towards smaller scales. By giving more weight to larger increments as p increases, structure functions of high order become increasingly dominated by the intermittent nature of the cascade and the exponents ζ_p deviate significantly from $p/3$ (see Anselmetti et al. 2001, for a review).

However, computing high order ($p > 4$) structure functions is demanding because the number of events in high-value bins of the PDF is intrinsically small (although much larger than Gaussian), and for the integral in Eq. (3) to converge requires large datasets (see Section 7.4). Another practical difficulty is that the expected scaling of the structure functions holds only in the inertial range of scales, which may be too small to achieve good accuracy. As a matter of fact, plotting $S_p(\ell)$ against $S_3(l)$ instead of ℓ extends significantly the range of $S_p(\ell)$ available for a power-law fit (Benzi et al. 1993). This property, called Extended-Self-Similarity (ESS), has been verified in many turbulent experiments at Reynolds numbers from 20 to 5000 (Arneodo et al. 1996). Expressing $S_p(\ell)$ as a function of $S_3(l)$ gives

$$S_p(\ell) \propto S_3(l)^{\tilde{\zeta}_p}, \quad \tilde{\zeta}_p = \zeta_p / \zeta_3. \quad (4)$$

making apparent the fact that the derived exponents are relative to ζ_3 . In incompressible turbulence, $\zeta_3 = 1$ is an exact result (proved only for the longitudinal velocity increments), and ESS thus provides the values of ζ_p . However, in strongly supersonic turbulence, such as in molecular clouds, $\zeta_3 \approx 1.3$ or higher (Kritsuk et al. 2007; Hily-Blant et al. 2008); yet, the ESS property still applies in such flows. In what follows, ESS is used to measure the structure function exponents.

Characterizing the turbulent cascade requires structure functions of high orders to be computed which in turn implies velocity increments at small scales to be measured with highest accuracy. In the context of molecular clouds observations, this translates into our ability to measure the CV from a spectrum. From Eq. (1), the uncertainty on the CV calculation is

$$\delta CV = \frac{\sigma_T}{W} \sqrt{\sum_{i=1}^N v_i^2 + N \times CV^2} = \sqrt{N} \frac{\sigma_T}{W} \sqrt{\langle v^2 \rangle + CV^2} \quad (5)$$

where σ_T is the spectrum noise (rms based on free-of-signal channels) and with v_i and N as in Eq. (1). The above expression assumes that the noise on the intensity is homogeneous for a given spectrum, and that the uncertainty on v_i is negligible. **While the second assumption is very well fulfilled, the noise on observational data is more complex than a white noise. However, a detailed formalism of the noise is out of the scope of this paper, and such a refinement is not necessary to recognize the impact of the methods presented.** This expression is the starting point of our motivations to develop a new approach to estimate CV. The expression on the right of Eq. (5) shows that both N and σ_T contribute to the uncertainty on the CV values. Optimizing the signal-to-noise ratio on the centroid velocity measurement may thus be achieved by acting on either two or both parameters, while taking into account the possibility of having multiple-component spectral lines with extended line wings.

A first approach is to center the spectra based on the local velocity of the cloud, which is done in `Unveil.jl`, but not enough. In what follows, we present two, complementary and fully independent approaches to measure the CV.

3. Test data set

3.1. Simulated clouds

In order to test `Unveil.jl` on realistic data sets, we built simulated 3D PPV cubes from MHD simulations. To this aim, we used the 256³ simulations of Cho & Lazarian (2003) from the CATS database (Burkhart et al. 2020). We only present the post-processing performed on these simulations and refer the readers to these papers for more details on the simulations themselves. The three-dimensional, density $\tilde{\rho}$ and velocity $\tilde{v}$ are initially dimensionless. Each simulation is identified by its sonic and Alfvénic Mach numbers, $\mathcal{M}_S$ and $\mathcal{M}_A$. To bring the cubes to physical units, three quantities must be fixed; we chose to define the gas kinetic temperature, T_0 , the average density of H nuclei, n_0 , and the mass $\mathcal{M}$ of the cloud. In what follows, we have used the mildly supersonic and sub-Alfvénic simulation referenced as s2a07 in the CATS database, for which $\mathcal{M}_S = 2$ and $\mathcal{M}_A = 0.7$. The cube was scaled with $n_0 = 300 \text{ cm}^{-3}$, $T_0 = 20 \text{ K}$, and $\mathcal{M} = 30 M_\odot$. The dimensionless mass density $\tilde{\rho}$ is then scaled to physical units as $\rho = n_0 \mu m_a \tilde{\rho}$, while the velocity follows from $v = c_s \mathcal{M}_S \tilde{v}$. The total size of the cube is $L = 1.4 \text{ pc} (\mathcal{M}/30 M_\odot)^{1/3} (n_0/300 \text{ cm}^{-3})^{-1/3}$. Other relevant quantities can be obtained: the velocity dispersion is $\sigma_v = c_s \mathcal{M}_S = 0.68 \text{ km s}^{-1}$, the thermal-to-magnetic pressures ratio is $\beta = 0.25$, the average magnetic fields intensity is $9.4 \mu \text{G}$, and the Alfvén speed is 1.0 km s^{-1} . The spatial resolution of the simulations is thus 55 mpc, or 3'' at 350 pc (the typical distance of the Polaris molecular cloud).

To construct ¹²CO spectra, we made highly simplifying assumptions on the abundance of CO in the simulation, since our primary aim is to construct data cubes with sufficiently complex spectra covering a wide range of line shapes and spatial variations generated by a turbulent flow. Although we do not perform chemical calculations, our prescription for the abundance of CO follows from the results of state-of-art physico-chemical models (Gong et al. 2017). The details of these calculations will be presented in a forthcoming paper (Perroni et al) and we provide here only the basic methodology. The CO abundance is parameterized based on the local visual extinction at each point in the simulation cube; this choice is motivated by the results of Gong et al. (2017), and was also tested on detailed isobaric models using the Meudon PDR code¹. **In the following, changed the widetilde above A_v into the underline to follow the same notation as for X**

$$\underline{X} = \underline{X}_{\min} + \frac{1}{2} \times (\underline{X}_{\max} - \underline{X}_{\min}) \times \frac{1}{2} \left[1 + \tanh \left(\frac{\underline{A}_v - b}{\Delta} \right) \right] \quad (6)$$

where $\underline{X} = \log_{10} X_{\text{CO}}$, and $\underline{A}_v = \log_{10} A_v$, and the four parameters $\underline{X}_{\min}$, $\underline{X}_{\max}$, b and Δ are the minimum and maximum abundance, the location and width of the C/CO transition in $\log_{10} A_v$, respectively. We use a fixed value of $\underline{X}_{\max} = -4$, while the other three parameters depend primarily on the density and cosmic-ray ionization rate, and to a lesser extent also on the metallicity and kinetic temperature. The spectra are then calculated by integrated the radiative transfer equation along the line-of-sight,

¹ Publicly available at <https://pdr.obspm.fr/>.

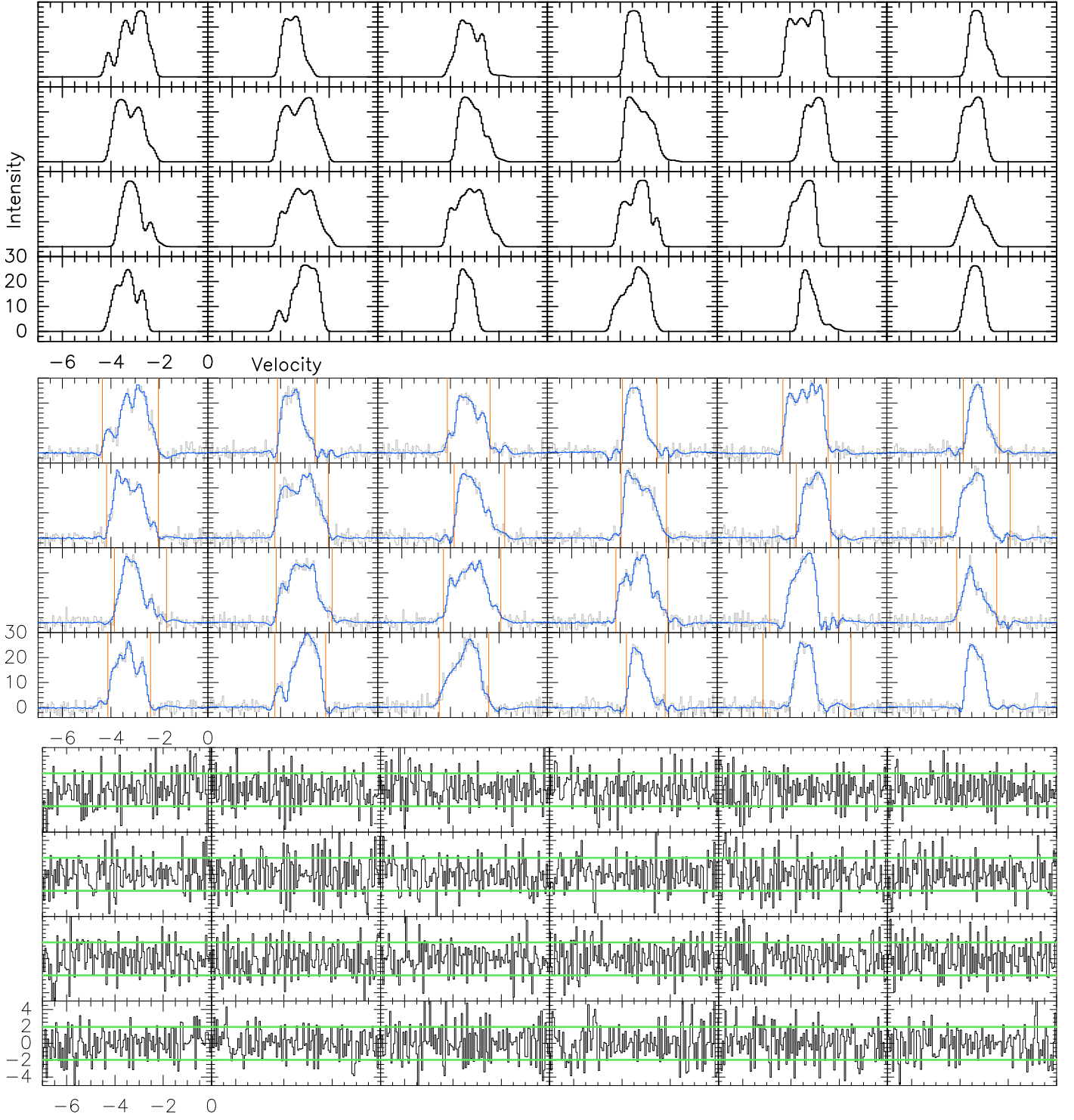


Fig. 1. *Top:* a random sample of noise-free simulated spectra based on the hydrodynamical simulation of [Cho & Lazarian \(2003\)](#); [Burkhart et al. \(2020\)](#). The velocity is in km s^{-1} , and the intensity in K (see also Fig. B.1). *Middle:* same spectra, from a cube with a SNR of 14 (gray), and from the PCA-reconstructed cube with $N_{\text{opt}}=19$ in blue. For each spectrum, the spectral window found by SWO is delineated with vertical orange lines. If none, SWO did not. *Bottom:* Residuals from PCA (original cube minus PCA reconstructed cube). The green lines are the added RMS equal to 1.95K.

with the CMB black-body radiation as a background. The contribution of each cell of size dz to the opacity of the $^{12}\text{CO}(1-0)$ transition, in velocity channel V , is given by

$$d\tau(x, y, z, V) = \frac{3c^3 A_{ul}}{8\pi\nu^3} \frac{1 - e^{-5.5/T_{\text{ex}}}}{Q(T_{\text{ex}})} \times n_{\text{CO}}(x, y, z, V) \Phi(x, y, z, V) \times dz \quad (7)$$

where $A_{ul} = 7.203 \times 10^{-8} \text{s}^{-1}$ is the coefficient for spontaneous radiative decay of the $J = 1 - 0$ transition, $Q(T_{\text{ex}})$ is the partition function, and T_{ex} the excitation temperature; the latter is assumed equal to the gas temperature, which is constant in these simulations. Finally, $\Phi(x, y, z, V)$ is the intrinsic line profile, taken to be a thermally-broadened Gaussian profile centered at velocity $v(x, y, z)$. The specific intensity is then converted into a radiation temperature using $I_\nu = 2k_B T \nu^2 / c^2$. The CMB is subtracted to the emergent spectra to mimic the ON-OFF procedure at millimeter wavelengths. The obtained noise-free PPV cube may then be added Gaussian noise. Since our purpose is primarily to benchmark our CV calculation procedure, we do not convolve the cube by a telescope beam. A random sample of noise-free spectra is shown in Fig. 1 (top panel). A full map is shown in Fig. B.1. As may be seen, a range of spectral line shapes is obtained, with well separated multiple or overlapping components, and also weak line wings. However, the line intensity are too strong by a factor 5 compared to the Polaris field.

3.2. Observations

The Unveil.jl pipeline is also applied to a real hyperspectral datacube, namely a map of the $^{12}\text{CO}(1-0)$ emission of the dense region in the Polaris Molecular Cloud (Hily-Blant & Falgarone 2007). The map covers $0.4^\circ \times 0.2^\circ$ with a $22''$ HPBW, and a 1σ noise of 0.40 K per spectral channel of width 0.055km s^{-1} . The typical SNR is ≈ 10 (peak average on the data cube). The cloud distance was recently revised upwards, being $385_{-22}^{+20} \text{pc}$ (Zucker et al. 2020). The region follows the brightest filament in thermal dust emission in the entire cloud (Miville-Deschênes et al. 2010), which harbors the only dense cores in the entire, 30 pc, cloud (Ward-Thompson et al. 2010; Wagle et al. 2015).

A statistical analysis of the turbulent velocity field in the Polaris Flare was already conducted with a method similar to the SWO approach Hily-Blant et al. (2008). We will thus compare the results obtained previously to those from the new PCA and SWO methods.

4. Spectral window optimization

To improve the accuracy on a CV, one way is to make N as small as possible that is, compute the first order moment on a window that is optimized according to some criteria. This is the approach followed by Pety & Falgarone (2003) who consider the signal-to-noise ratio on the integrated area as the figure of merit to optimize N :

$$SNR = \frac{\sum_{i=p}^{p+N-1} T(v_i)}{\sigma \sqrt{N}} \quad (8)$$

where p is the index of the first channel. The SNR is computed for varying p and N and reaches a maximum for the optimized window.

In view of handling multi-components, or multi-peaks spectra, and to have a more robust optimization criteria, we have devised a new approach which will be refereed to as SWO, for Spectral Window Optimization. To determine the best window, we define the normalized cumulative intensity, noted C_m , and compute it in two directions, starting from either end of the ve-

locity range:

$$C_m^+ = \frac{\sum_{i=1}^m T_i}{\sqrt{m} \sigma_m}, \quad C_m^- = \frac{\sum_{i=N-m}^N T_i}{\sqrt{m} \sigma_m}, \quad (9)$$

where σ_m is the rms of the intensity computed in the running, increasing window of size $m = 2 \dots N$ channels. The typical run of either cumulative with increasing m is to remain close to zero until m reaches to the beginning of the spectral line, at which point C grows; upon exiting the line, C then decreases (see Fig. 3). The locations of the two maxima of C^+ and C^- provide the optimized window. The choice of the denominator in Eq. (9) is empirical based on various trials, and was found to be robust against a wide variety of line profiles.

5. The PCA method

5.1. Overview and aims

The Principal Component Analysis (PCA) is a mathematical tool employed in this paper to reduce the influence of noise on our spectral line data cube. Although PCA is already utilized in the denoising of 2D images (Weston et al. 2004), this work extends its application to these spectral line data cubes, providing a novel approach for noise reduction in these datasets.

PCA consists in reprojecting the data on a new orthonormal basis. The central feature of the PCA is that the new basis vectors are sorted in such a way that the variance of the data on the first new vector is maximum, and decreases with increasing basis vectors. It is thus said that the new basis vectors are sorted with decreasing variance, the latter being the amount of significant information in a statistical sense.

Mathematically, it can be shown that the principal components are the solutions of an eigenvalue problem, the eigenvalues being the amount of variance carried by each eigenvector. The first eigenvector is the linear combination of the original basis vectors that condenses most of the variance. On the opposite, the last eigenvectors contribute to little variance that is, they carry very little information, or essentially noise. A key feature of the PCA is that the eigenvectors form an orthogonal basis, ensuring that the decomposition is non-degenerate, like the Fourier or some wavelet decompositions. This is in contrast with Gaussian-based approaches to the decomposition of line profiles.

PCA is popular in many fields of science, as it provides a fast and reversible way of sorting the data according to the amount of significant information. In radio astronomy, it was first introduced by Heyer & Schloerb (1997), who used the PCA to determine the slope of the linewidth-size relation in molecular clouds. The PCA is also used in astrochemical studies as a way to identify chemical species with similar spatial distributions and thus group them into consistent, chemically related, sets of species (Ungerechts et al. 1997; Gratier et al. 2017). PCA is also popular in data processing where it is used for data compression. The basic idea is that any eigenvector beyond a certain rank contributes negligibly to the variance and can thus be considered meaningless. This is the feature of the PCA used in the present work, where it is applied to observed PPV cubes to reconstruct artificial data cubes that contain all the spectral information down to some threshold. The resulting cube has a lower value of σ_T in Eq. (5).

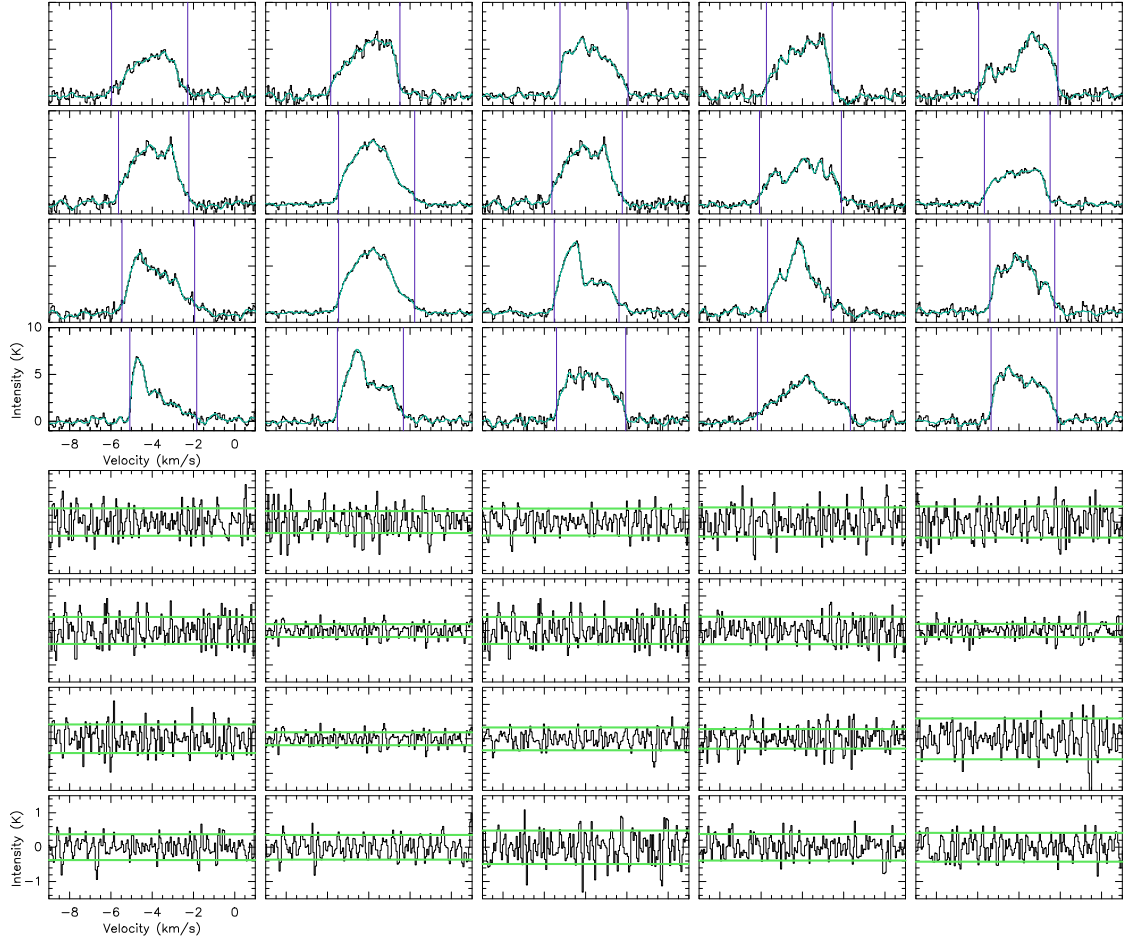


Fig. 2. *Top* : Randomly picked spectra on Polaris observational data. PCA-reconstructed spectra with $N_{\text{opt}}=29$ components are shown in green, while the vertical lines in each panel locate the spectral window obtained by SWO. *Bottom* : Residual of the above spectra (original cube minus PCA reconstructed cube with N_{opt}). The green lines are the intensity dispersions computed on noise channels of the given spectra.

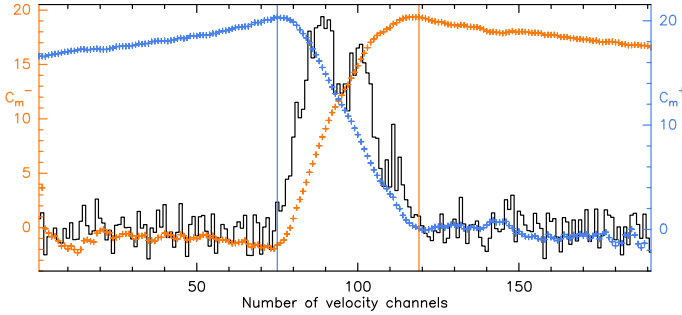


Fig. 3. A simulated spectra of the SNR=14 simulated cube (see Section 3.1) processed by SWO. Blue and orange markers are values of C_m^+ and C_m^- respectively. Vertical lines are the positions of their maxima.

Our work addresses a significant challenge in Principal Component Analysis: the accurate recovery of the appropriate amount of variance. We developed our method based on universally applicable and user-friendly criteria, aiming to maximize the ease of use.

5.2. Implementation

Our implementation of the PCA to PPV cubes follows Heyer & Schloerb (1997): the original cube of size (N_x, N_y, N) is reshaped into a rank-2 matrix of size $(N_x \times N_y, N)$, and the new eigenvec-

tors are linear combinations of the N velocity channels. The new orthonormal vectors are noted $\varpi_i, i = 1, \dots, N$. This eigenvalue problem is solved using the Julia package *MultivariateStats.jl*², using a singular value decomposition. After the N eigenvectors ϖ_i have been determined, the intensity can be reprojected on the new orthonormal basis.

In practice, ϖ_1 is the average spectrum of the input data cube that is, $\varpi_1 = \sum_k \langle T \rangle_k V_k$. Therefore, the values in the first channel map of the reprojected cube tell how much the original spectrum at each position resembles the average spectrum. The intensity in the second channel map gives the largest deviations to the average spectrum, the third channel the deviation from the previous two, etc. An example of application of the PCA to a simulated data cube is shown in Fig. A.2. The first eigenimage $\tilde{T}_i$ shows how much each spectrum resembles the average one, and the channel map is quite uniform compared to the next channels. Reconstructing the cube with the N eigenvectors reproduces strictly the input cube.

5.3. Optimization criteria

The basic idea of using the PCA for data compression is to reconstruct a data set using $N_{\text{opt}} < N$ principal components, considering that the eigenchannels $i > N_{\text{opt}}$ only contribute to noise. Applied to PPV cubes, the reconstructed spectra thus contain

² <https://github.com/JuliaStats/MultivariateStats.jl>

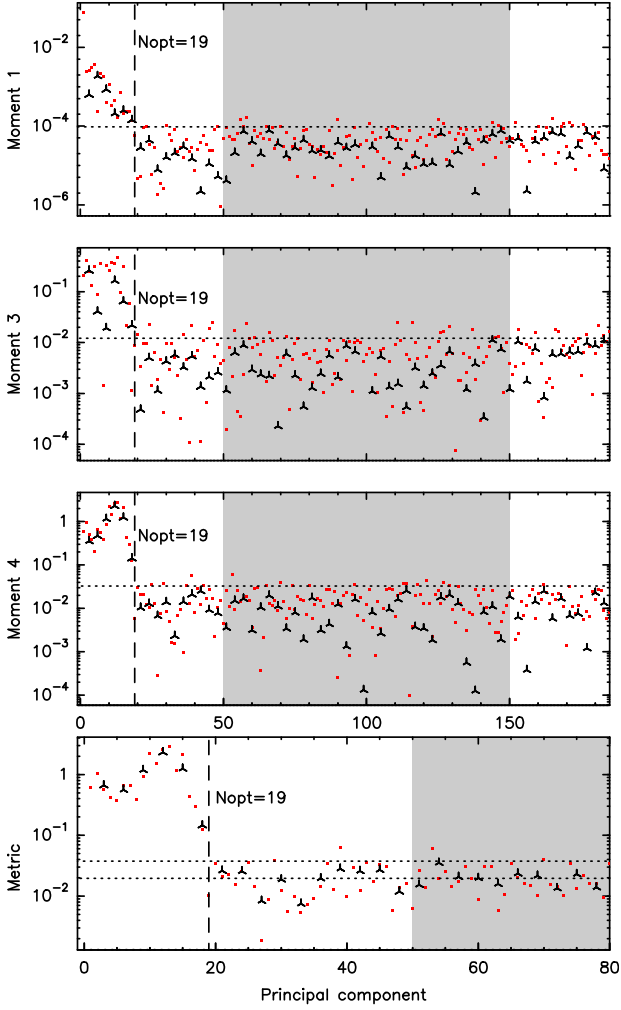


Fig. 4. Calculation of the metric φ_i (see Eq. (10)) for each principal component $1, \dots, N$ for a simulated $^{12}\text{CO}(1-0)$ cube with a SNR of 12. Top to bottom: absolute values of the moments μ_i , γ_i , and $\kappa_i - 3$. In each panel, the red points show all the principal components, while the black symbols show averaged values computed every three components. The grey area shows the range of components used to compute $\langle \varphi \rangle_m$ and σ_m used in the determination of N_{opt} according to the criteria of Eq. (11). For each moment, the horizontal dotted line shows $\langle \varphi \rangle_m + 3\sigma_m$ and the vertical dotted line indicates the value of N_{opt} . Bottom panel: the resulting metric shown on a smaller range. The horizontal dotted line shows the values of $\langle \varphi \rangle_m$ and $\langle \varphi \rangle_m + 3\sigma_m$ (see Section 5.3).

less variance than the original ones, while still carrying all the meaningful spectral information. The determination of N_{opt} is thus a critical step. Despite the very wide use of PCA in data science, there is no general criteria that may be applied blindly to data sets of any kind. The simplest, and often quoted, criteria would be to retain the principal components ϖ_i with eigenvalues $\lambda_i \geq \lambda_{\min}$: thus N_{opt} is such that $\lambda_{N_{\text{opt}}} \geq \lambda_{\min}$ and $\lambda_{N_{\text{opt}}+1} < \lambda_{\min}$. However, we found this method was not sufficient for our case, leading to numerous artifacts and loss of spectral line features such as the line wings. Our approach to determine N_{opt} is based on the expected properties of the noise in the data. We thus introduce a criteria based on the statistical properties of the signal in each eigenimage, rather than on the amount of variance reproduced by a given eigenvector (its eigenvalue). Assuming that the noise in the input data cube is Gaussian, the proposed criteria is that the signal in an eigenimage $i > N_{\text{opt}}$ is a normally distributed, noise-like signal. For each eigenimage i , we thus com-

pute the following metric

$$\varphi_i = \sqrt{\mu_i^2 + \gamma_i^2 + (\kappa_i - 3)^2}, \quad (10)$$

where μ_i , γ_i , and κ_i are the moments of order 1, 3, and 4 of the signal in the eigenimage $\tilde{T}_i(x, y)$. For strictly normally distributed, centered, values, the mean μ_i and skewness γ_i are zero, and the kurtosis $\kappa_i = 3$. Thus, a strictly normally distributed signal has $\varphi_i = 0$. In practice, the metric is never strictly zero; yet, beyond a certain principal component, the moments and the metric fluctuates around a low value. We thus compute for each moment M a mean $\langle M \rangle$ and rms σ_M in a given range (grey area in Fig. 4): this range is tuned for each dataset, and may not reach the upper bound N because of non-homogeneous noise in the original data (see Fig. C.2). The value of N_{opt} for moment M is such that

$$M_{N_{\text{opt}}-1} > \langle M \rangle + 3\sigma_M. \quad (11)$$

A similar calculation is done for the metric. In the process, the individual values of M_i and φ_i are smoothed with a running square box of width 3: this allows to mitigate the fluctuations and to obtain a meaningful value of N_{opt} . Our procedure thus delivers four estimates of N_{opt} , one for each moment and one for the metric; although these were found to coincide in all our tests, they may not in some others. Nevertheless, the value of N_{opt} is a user-defined parameter on the basis of the metric analysis.

The entire procedure is summarized in Fig. 4, performed on a simulated cube to which a homogeneous noise was added, such that the average peak SNR along lines of sight is 12. On this case, the four values of N_{opt} are equal to 19. The behavior of the metric is typical, with high values in the principal components that contribute most to the variance, followed by a steep decrease beyond which the value fluctuates around a value two orders of magnitude smaller, associated with the principal components that essentially reproduce the original noise. The PC range used to compute the mean and rms was here set to 50 to 150, where they acquire an average of -1.2×10^{-5} , -7.2×10^{-4} and -1.8×10^{-3} respectively, with dispersions of 3.6×10^{-5} , 4.3×10^{-3} and 1.1×10^{-2} . An example of the eigenimages distribution shape is presented in Appendix A where we understand the impact of the noise and the signal.

6. Description of the pipeline

The `Unveil.jl` pipeline is written in the *Julia* programming language. This choice was motivated for the speed, ease of use, built-in GPU programming facilities, and growing community of *Julia*. For now, GPU multi-thread is not implemented in `Unveil.jl` but will be in a near future. The `Unveil.jl` module is not an officially referenced package by *Julia* because it will only be used by a small part of the *Julia* community, but can be found on Github³.

A minimum set of front-end functions have been defined that allow the basic operations to be performed in a transparent way: PCA, SWO, metric, CV, CVI, structure functions (and convergence) and their exponents. Each of these functions use an input file containing every parameters needed (for example, the path to the data cube fits). Templates for these input files can be generated automatically using a built-in function. The front-end functions rest upon back-end functions that are partitionned into seven modules

³ <https://github.com/Simon-Delcamp/Unveil.jl?tab=readme-ov-file>

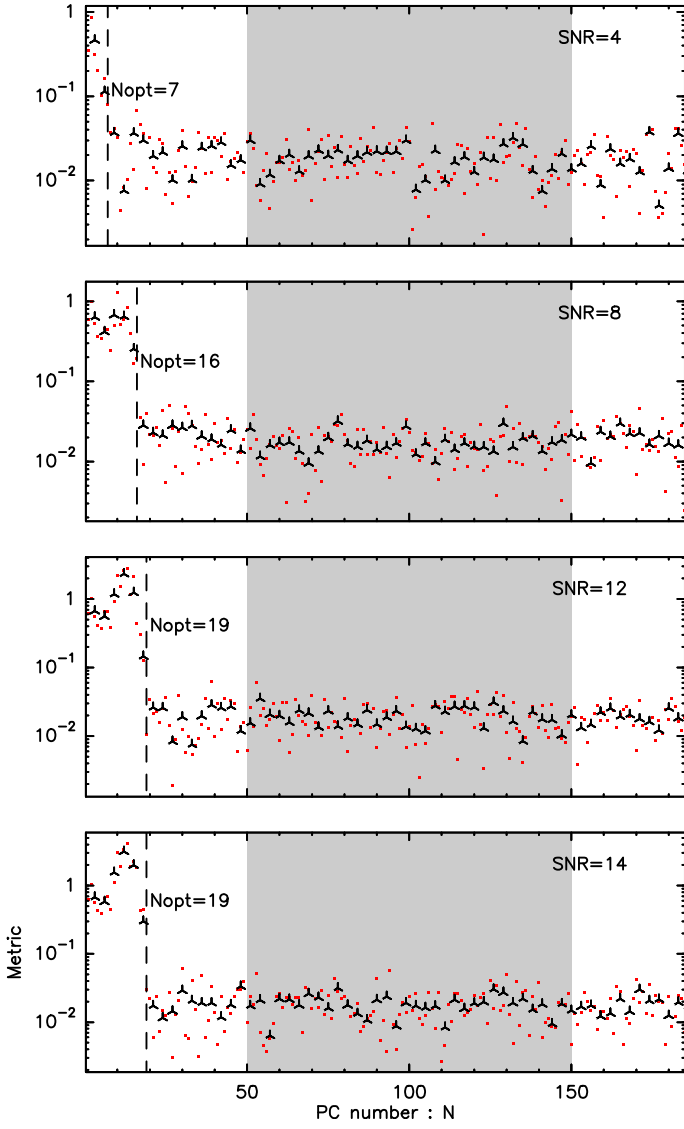


Fig. 5. Same as Fig. 4 for four instances of the **average peak** signal to noise ratio (SNR). Only the metric ϕ_i (see Eq. (10)) is shown here. From top to bottom: SNR=4, 8, 12, and 14, and the values of N_{opt} are 7, 16, 19 and 19 respectively.

depending on their general aims: initialization, graphic display, etc. For each of these functions (back- and front-end), a documentation is written, describing how and which variables to use.

The general structure of the `Unveil.jl` code is illustrated in Fig. 6. Overall, the `Unveil.jl` package consists of more than 2.6k rows. The seven front-end function of `Unveil.jl` use `ascii` files as inputs. The parameters are of multiple types and reflect the different parts of the pipeline. Three types of outputs are generated by `Unveil.jl`: data file in the form of `fits` files, data files in the form of simpler, `ascii` tables, and finally PDF plots. The output `fits` files use the same header as the input one, with modified entries if needed.

It is difficult to provide a quantitative value for the total execution time of the pipeline because several steps require inputs from the user on the basis of the results from previous front-end functions. One such instance is the convergence analysis: it is a feature of `Unveil.jl` to provide various plots and figures of merit, such as the metric of

Eq. (10) to help the user setting a value of N_{opt} , rather than fixing it automatically. Nevertheless, we can provide a typical value by summing the execution times for the basic steps of the pipeline. Using the `BenchmarkTools.jl` package⁴, and times⁵, the treatment of a cube with 15k spectra and 218 velocity channels is less than a minute (assuming every variables files are ready to use). In details, SWO takes ~ 7 s and PCA less than one second and the PCA converging analysis requires ~ 2 s; then, CV and CVI computations with 14 lags need ~ 10 s, and the structures functions for $p = 1$ to 6 ~ 5 s. We saw a more important dependence of the computing time with the size of the cube using the SWO method than the PCA one. The CVI computation is the more time-consuming process, while it is also highly memory dependent. Indeed, an array with every CVI should be created in memory, containing the differences at every lag, and every rotation. Because the number of rotations needed, at a given lag, is dependent on its size, a large map from which we would like to compute large lags will create a CVI cube much larger than a smaller map.

7. Code benchmark

In this Section, we present a series of tests of our pipeline. We applied the code `Unveil.jl` to a realistic simulation based on a real data cubes. Simulated cubes offer the opportunity to compare the results of the CV calculation applied to a cube with noise, to the **correct** result from the noise-free simulation and thus to characterize the impact of the SNR on the performances. **All SNR values given below are average peak SNR along lines of sight.**

7.1. Convergence of the PCA

Figure 5 shows the impact of the SNR on the metric and the derived value of N_{opt} , by using a single simulated cube to which a white noise of decreasing rms was added. As the SNR is increased, the value of N_{opt} rises, reflecting the fact that more components are needed to reproduce the signal. Indeed, as the rms decreases, the amount of statistically meaningful information increases and so does the number of principal components required to account for it. Nevertheless, the SNR study shows the convergence of N_{opt} with increasing SNR, here at 19 for a SNR of 12 (see also Fig. 4).

The noise rms of the input cubes are 10.36, 3.76, 2.32, and 1.96 K. These estimations are computed on the noisy cube as if they were raw data obtained after observations. Thus, the maximum averaged signal is different from each cube, which explains the non-linear relation between rms given above. Reconstructing the cubes with N_{opt} , their noise RMS are now respectively 0.09, 0.14, 0.31, and 0.30K. For the following analysis, all results will be shown with the respective N_{opt} of each cube.

7.2. Comparison of PCA and SWO: CV computations

We applied the PCA and SWO methods on each of the noisy cubes. For the sake of clarity, the output cubes will be referred to as the "reconstructed" and "treated" cubes, for PCA and SWO respectively. The noise-free cube is not treated nor reconstructed

⁴ <https://github.com/JuliaCI/BenchmarkTools.jl>

⁵ Because of the just-in-time compilation, a script is longer at its first use in Julia than its second (Bezanson et al. 2017). The reported execution times thus do not refer to these initialization runs.

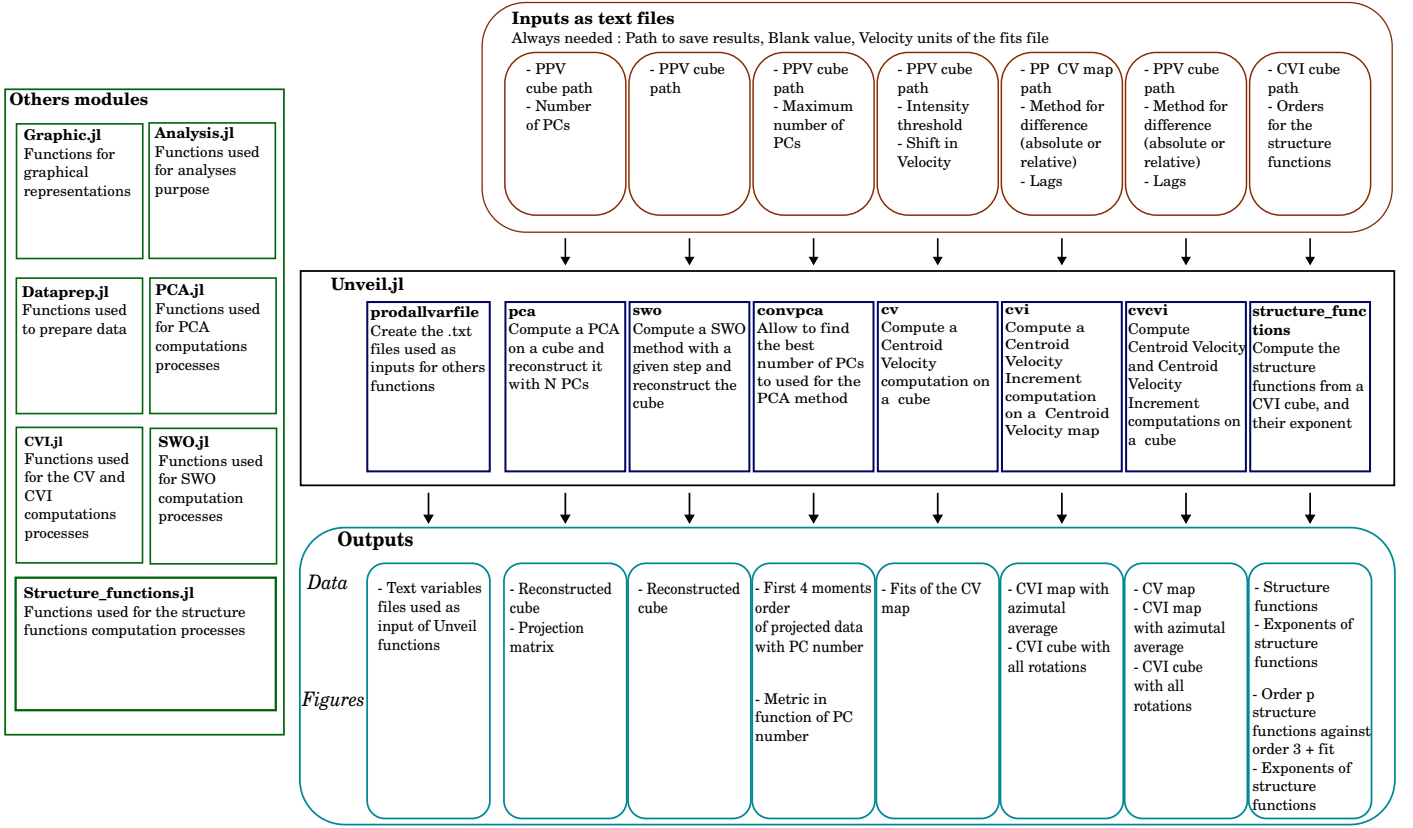


Fig. 6. Structuration of the code `Unveil.jl`. The main program is in blue, separated on multiple functions, each using a dedicated input txt file in orange. Modules with dedicated specific function called by the main program function are in green. The given outputs are in light blue, separated between each main program function and by their type (Data or figures).

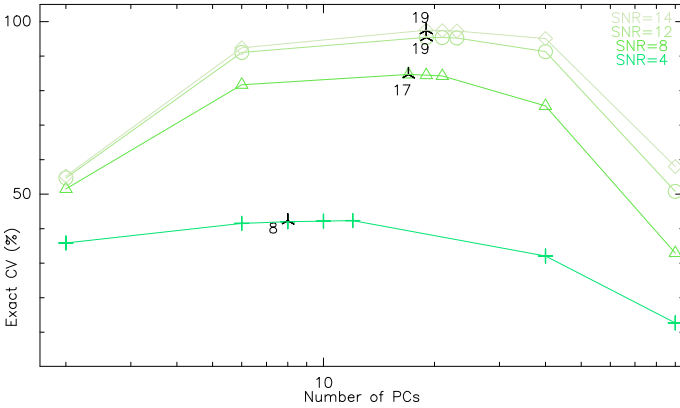


Fig. 7. Percentages of CV that differ by less than δv from the target values, as a function of the number of PC. Four cubes with increasing SNR (4, 8, 12, and 14) are shown with colours. The black markers corresponds to N_{opt} for each cube.

by these methods and is used as a reference; the CV computed on the noise-free cube are the target values. The CV computation is thus performed on the treated and reconstructed cubes, and on the noise-free cube. Furthermore, we also compute the CV on the noisy cube without any method; we will refer to it as the raw cube.

By comparing the CV computed on cubes reconstructed with different varying number of principal components N , we compute the percentage of correct CVs, which are defined as those which differ from the target values by less than one velocity

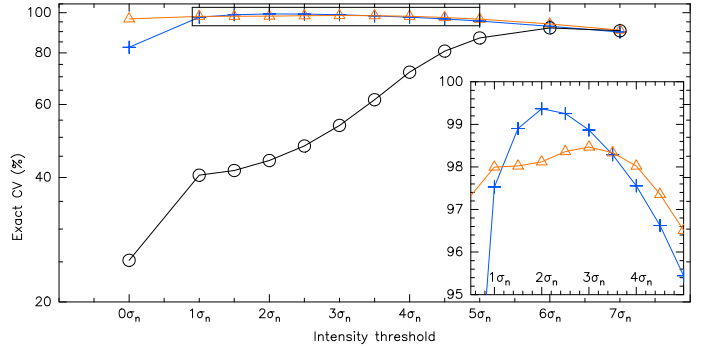


Fig. 8. *Top*: Percentage of **correct** CV from the SNR=14 cube, as a function of the intensity threshold applied to three different cubes: raw cube (black), PCA-reconstructed (blue) and SWO-treated in orange. Each intensity threshold is given in value of the input noise σ_n . A zoom of the contoured limits is given on the bottom right, centered on results from PCA and SWO.

channel δv . **By doing so, we consider that two values separated by less than the velocity resolution are not scientifically robust because of the intrinsic instrumental limitation.** The results are presented Fig. 7. Each level of SNR gives the same evolution: when increasing N , the percentage of **correct** CV first increases, reaching 99% for the least noisy cube. This number of **correct** CV then reaches a plateau which is followed by a decrease of **correct** CV down to value comparable to the initial one. The value in the plateau increases with the SNR; this is indeed true for the entire curves.

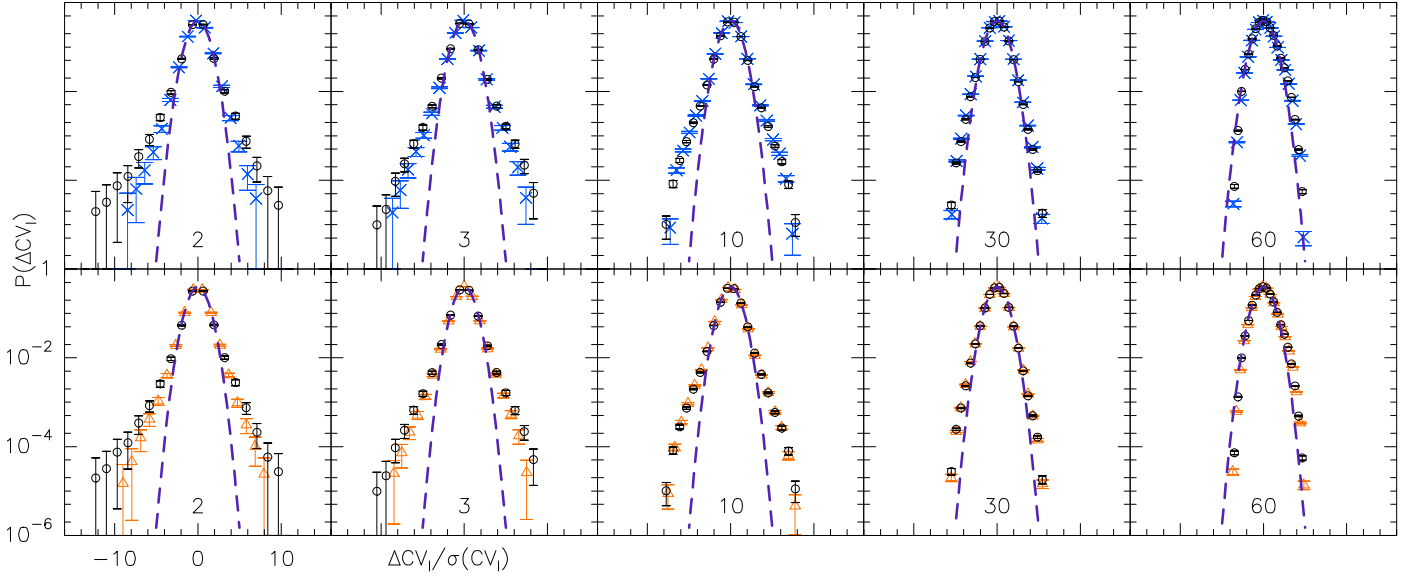


Fig. 9. Distributions of the CVI of the simulated cube at SNR=14 at multiple lags and for each methods. Lags are given in pixel units inside every panel. Each black distribution is coming from the noise-free cube. The first line correspond to results from PCA with $N_{\text{opt}}=19$ in blue crosses. The second is for SWO in orange triangles. A Gaussian of zero mean and dispersion unity is also shown in dashed purple curves. Every distribution is normalized at zero mean and at dispersion unity. Each bins contains at least 10 data points. Uncertainties on the bin numbers are statistical.

The reconstructed spectra from the SNR=14 cube, with $N_{\text{opt}}=19$, are presented at the center of Fig. 1. Close to the spectral line, negative artifacts are seen for almost every spectrum reconstructed by PCA. Nevertheless, these artifacts have an amplitude always smaller than the initial noise. The impact of these artifacts on the CV calculation was suppressed by applying an intensity threshold before the CV computations. The threshold affects only weakly the number of correct CVs, as may be seen in Fig. 8 for the cube with SNR=14. Similar results are obtained for the cubes with SNR=4, 8, and 12 (not shown). With no threshold, the percentage is always the lowest, by almost a factor 4 in the cube without treatment or reconstruction. The best performance is achieved with the PCA, $\sim 1\%$ higher than SWO, when using a threshold of $2\sigma_n$. The best percentage for SWO treatment is with a threshold of $3\sigma_n$. The performance on the raw cube converges towards the results of the PCA and SWO at high intensity threshold, higher than $6\sigma_n$. Although PCA reaches better performances, SWO is less sensitive to the threshold than PCA, due to the artifacts mentioned above. These are created by the intrinsic operations of PCA, adding negative and positive intensities for each new N used for the reconstruction. Nevertheless, the sensitivity of the PCA cube is also very limited as soon as a threshold at $1\sigma_n$ is applied. Unexpected evolution is obtained with the noisiest cube which is due to the too high noise in the data, which will be repercutated in the following sections.

7.3. CVI distributions

We now explore the performance of the PCA and SWO methods, and their differences, regarding the statistical properties of the CVI. The distributions of the CVI for each of them are shown in Fig. 9 and compared to the results using the noise-free simulation. For both methods, the typical behavior of intermittent turbulent flows is recovered, namely non-Gaussian wings towards small lags, and a Gaussian distribution at large ones (Anselmetti et al. 1984). The wings start to show up at $\pm 2\sigma$. Both methods underestimate the amplitude of the non-Gaussian wings at small lags. A slight asymmetry is noticeable at the largest lag $\ell = 60$

pixels, which is in the opposite sense as the usual asymmetry in increment distributions (Anselmetti et al. 2001), an effect that could be due to a too small simulation box to ensure that CVI at this lag belong to the inertial domain.

The wings of these distributions are markers of intermittent turbulent dissipation (Lis et al. 1996). It is thus of crucial importance to reproduce them correctly. To obtain a better quantitative comparison between the efficiency of the methods, we computed the second-moment order of these distributions (see Fig. B.2). For every cube, at lags $\ell \geq 7$ pixels, PCA recovers the exact dispersion (within the uncertainty of the velocity resolution). For the highest noise cube, SWO neither gives any correct value, with an order of magnitude of differences. SWO gives correct values within the $1\delta v$ uncertainty from the SNR=8 cube. Compared with SWO, PCA performed better in every configuration. Differences between PCA and SWO decrease with the lag or with the SNR increasing.

7.4. Effects on the structure functions

Velocity structures functions are at the heart of the turbulence statistical analysis. They are also the end product of the pipeline together with their scaling exponents. In particular, the detailed description of the small-scale behavior of the velocity field is encapsulated into the high-order structure functions. We thus explore the performance of the PCA and SWO on the determination of the structure functions.

We present in Fig. 10 the integrand of Eq. (3), for structure functions up to order $p = 6$ obtained from the SNR=14 cube. This Figure exemplifies the general behavior of the integrand: as p increases, values far from the mean (here equal to zero) are given increasing weight. In so doing, the scatter of the values also increases, reflecting the fact that values far from the mean are intrinsically rare, despite belonging to the non-Gaussian wings at the small lag ($\ell = 2$) shown here. This behavior applies equally to the noise-free cube, and to cubes analyzed with PCA or SWO. However, the two methods lead to differences with the targeted values reach one order of magni-

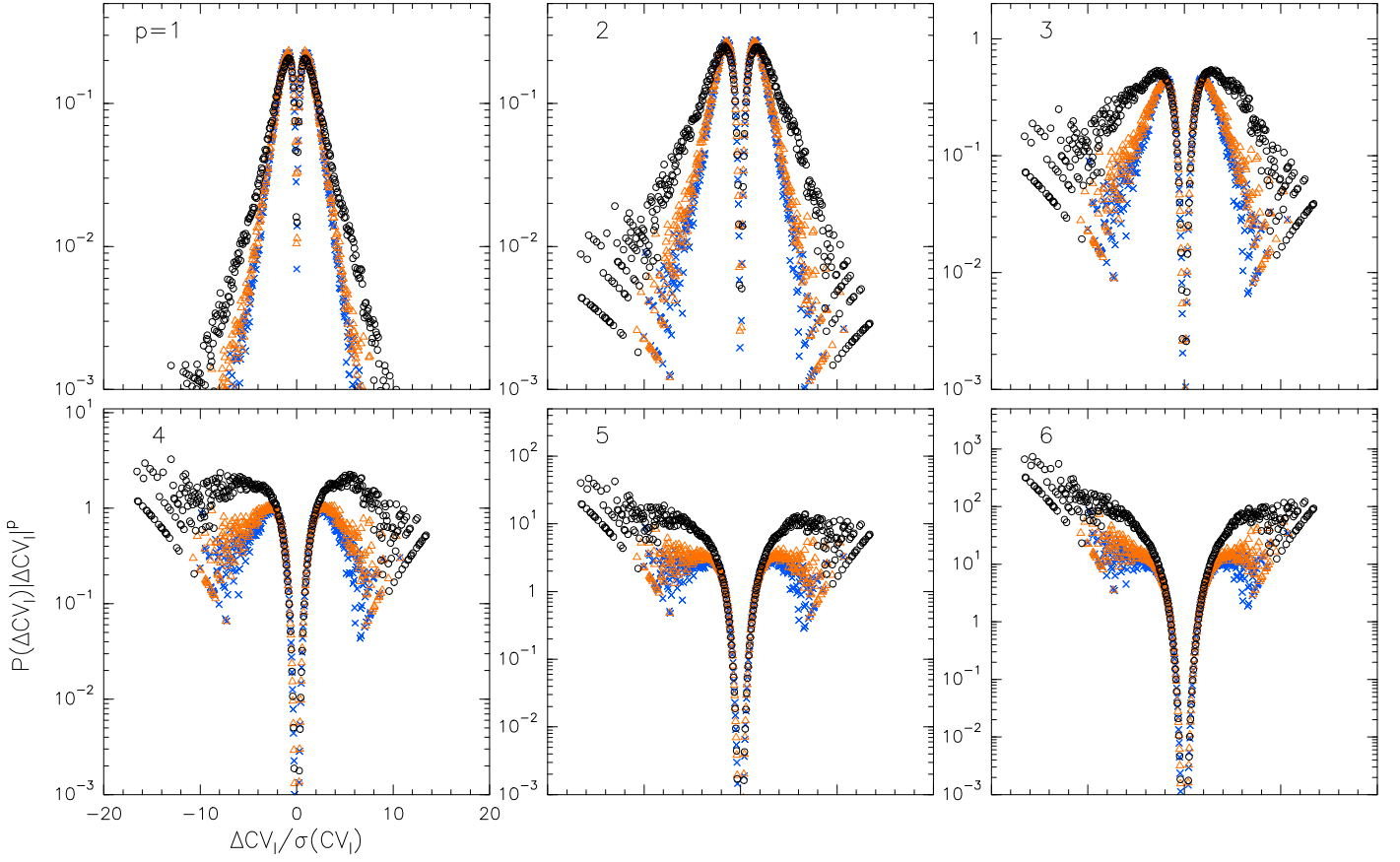


Fig. 10. Integrant of the velocity structure functions $S_p(\ell)$ at $\ell = 2$ pixels, for $p = 1$ to 6 (see Eq. (3)), computed from noise-free cubes (black circles), reconstructed by PCA (blue crosses), and SWO (orange triangles). Each panel is a different order, indicated in the top-left corner.

tude beyond $\pm 1\sigma$, with PCA at N_{opt} giving similar values than from SWO.

The $S_p(\ell)$ for $p = 1$ and 6, from the SNR=4 and SNR=14 and obtained with PCA at N_{opt} and SWO, are compared to the target structure functions in Fig. 11. At a SNR of 4, it is not possible to recover the structure functions with either method, although PCA does much better than SWO. At a SNR of 14, the $p = 1$ is very well reproduced by either method, but at small lags $\ell \leq 6$ pixels where the difference is a factor 1.3 compared to the target values. At order $p = 6$, PCA performs better than SWO at small lags and converges very close to the target function.

Finally, we explore the performance of the two methods on their ability to recover the scaling exponents of the structure functions. However, it must be recognized that the limited spatial range of the simulation is an intrinsic limitation on the measurement of the exponents, since the inertial range is at most one decade. This is why we have used the ESS assumption throughout. The fits of the structure functions are shown for the SNR=4 and 14 cubes in Fig. B.3. For the noisiest cube (SNR=4), SWO is outperformed by PCA which recovers the noise-free values; however, since the structure functions are not well reproduced even by PCA, this result must be interpreted primarily as a demonstration of the better performance of PCA over SWO. SWO gives already better results at SNR=8 for the highest lag than SNR=4, with differences of a magnitude but reproducing the global shape. At the highest SNR, both methods give similar structure ESS $S_p(\ell)$.

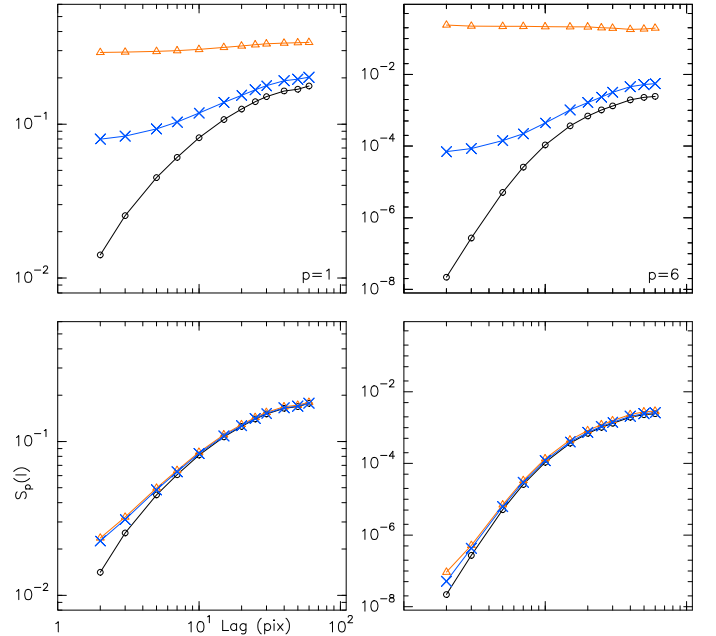


Fig. 11. $S_p(\ell)$ plotted against the lags for SNR=4 and SNR=14 cubes (top and bottom lines respectively): from the noise-free in black, SWO in orange triangles, PCA with N_{opt} in blue crosses.

7.5. Effect on the exponents of the structure functions

The resulting ESS exponents $\tilde{\zeta}_p = \zeta_p/\zeta_3$ up to $p = 6$, obtained from fitting the $S_p(\ell)$ from lag $\ell = 3$ to 40 pixels, are shown in

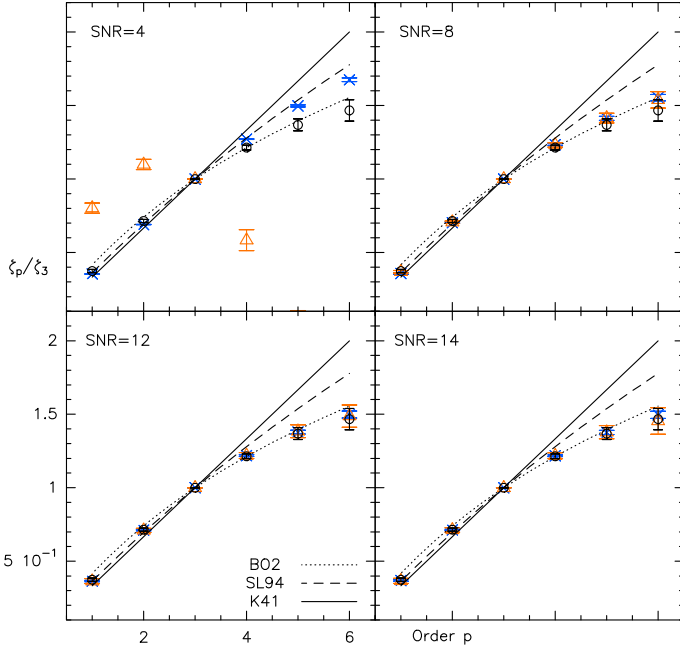


Fig. 12. Exponents $\tilde{\zeta}_p = \zeta_p/\zeta_3$ of the velocity structures functions computed from simulated cubes of different SNR treated and reconstructed with SWO and PCA. The color code is the same as Fig. 11. Lines are different turbulent dissipation models, with K41 for Kolmogorov (1941), SL94 for She & Lévéque (1994) and BO2 for Boldyrev et al. (2002). For SWO at SNR=4, values at high p are not shown.

Fig. 12. Not surprisingly, SWO is discarded for the SNR=4 cube. For this cube, the low-order exponents are well reproduced by PCA, while the higher ones are not, reflecting the discrepancy at small lags where the intermittent behavior is largest and mostly concentrated at high orders. For the cubes with better SNR, both methods converge towards the target exponents within the uncertainties. The differences between the PCA and SWO methods provide an estimate of the uncertainties attached to the methods. It is also found that, for a given method, the differences with the target values are very comparable for SNR=12 and 14 (see Fig. B.4). In addition, the PCA method is robust against all these SNR, with reproducible differences from SNR=4 to 14, while SWO presents exponents that change drastically between SNR=4 and 8, and are almost constant at higher SNR. Finally, the exponents (target values and PCA or SWO) are closer to the theoretical predictions for a shock-dominated, MHD turbulence (Boldyrev et al. 2002) than to the mildly incompressible, hydrodynamical case She & Lévéque (1994); Porter et al. (2002).

8. Application to the Polaris Molecular Cloud

8.1. Results of PCA and SWO processes

The new approaches of CV calculation developed in `Unveil.jl` are now compared to the method used previously in Hily-Blant et al. (2008), and apply the complete pipeline to the Polaris data cube. The PCA was thus applied to the Polaris data. The eigenimages shown in Fig. C.1 share the general behavior discussed previously, namely the first channel is the average emission and subsequent channels the differences to the average, in decreasing order. The convergence study of the PCA leads to an optimal number of components $N_{\text{opt}}=29$. The corresponding metric is presented in Fig. C.2. It is similar to the metric obtained with simulated data although there is an increase of φ_i for channels

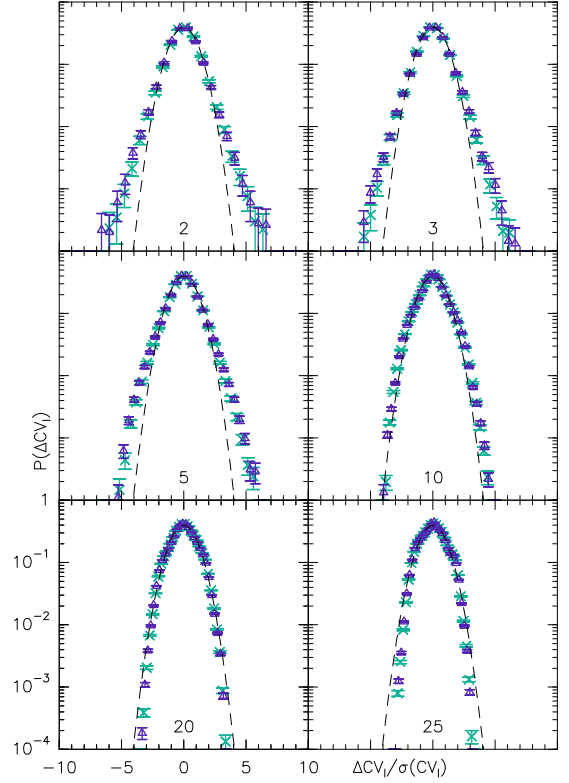


Fig. 13. CVI distributions of the Polaris observational data for 4 lags. Lags are given in pixels at the bottom of each subplot. In green crosses are results from PCA, in purple triangles from SWO. In black is a gaussian distribution with zero mean and of dispersion unity. The CVI are computed as relative differences. Bins with less than 10 points are deleted.

$i > 150$; this is due to the spatially inhomogeneous noise in the map, which was obtained as a collection of smaller sub-maps, with different noise (see Fig. C.3 for the noise map).

A set of 20 randomly chosen spectra are presented Fig. 2 together with their reconstructed spectra using PCA with $N_{\text{opt}}=29$. As may be seen, the reconstructed spectra reconstruct very well the observed spectra together with noisy features of the baseline. The RMS of the reconstructed cube with N_{opt} is 0.23K, almost twice smaller than the original value of 0.41 K. For the CV computation, we used an intensity threshold of $2\sigma_n$ for PCA and $3\sigma_n$ for SWO based on the above study on simulated cubes, σ_n being computed from the original cube. The CV maps obtained from each methods are described in Fig. C.4 and compared in Fig. C.5. Figure 13 presents the derived relative CVI distributions. The two methods give the same results at every lags, with non-Gaussian wings observed for lags $\ell \leq 5$ pixels.

The velocity structure function integrands, from both methods, are plotted in Fig. 14. They are very similar although that from PCA shows less scatter, especially at orders $p > 1$; for instance, the result from SWO presents an outlier at $\Delta CV_l \sim -7\text{km s}^{-1}$ for each order p . Both methods lead to decreasing integrands, thus converging structure functions, for $p < 5$; for $p = 5$, the PCA flattens after reaching a maximum while SWO is more scattered; at $p = 6$, no maximum is obtained. We thus find that the structure functions of $p \leq 4$ converge, while they do not for $p = 6$; $p = 5$ is marginally converging for PCA only. Nevertheless, in what follows we compute the structures functions up to $p = 6$ to enable a comparison with Hily-Blant et al. (2008).

The exponents of the velocity structure are obtained using the ESS approach, which lead to a power-law scaling from $\ell = 2$

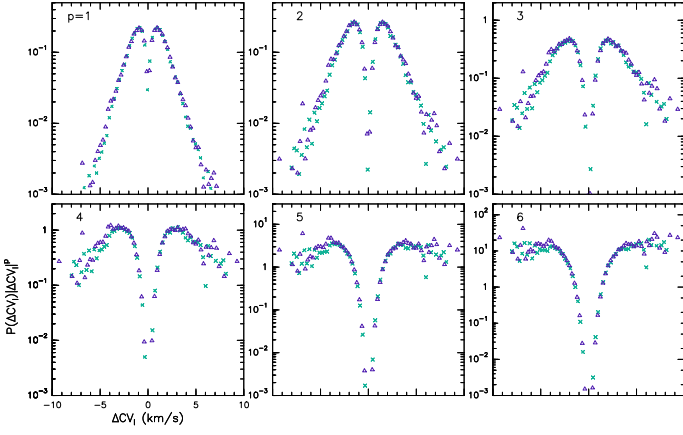


Fig. 14. Integrand of the structure functions at $\ell = 2$ pixels (see Eq. (3)) of the Polaris data cube reconstructed with N_{opt} by PCA (green crosses), and treated by SWO (purple triangles). The corresponding order p is written in the top-left corner of each panel.

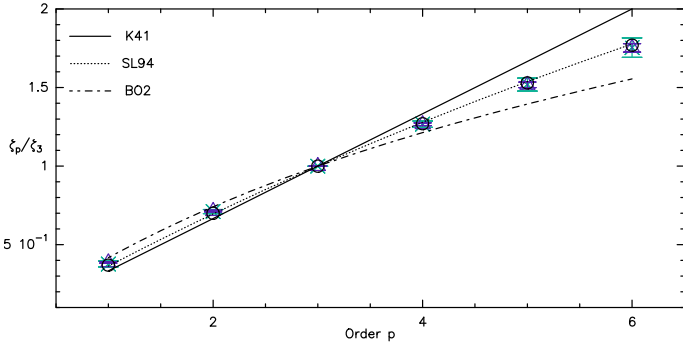


Fig. 15. Exponents $\tilde{\zeta}_p = \zeta_p / \zeta_3$ of the velocity structure functions for the Polaris cloud, from the PCA (with $N_{\text{opt}}=29$, green crosses) and SWO (purple triangles) methods. The black circles are taken from Hily-Blant et al. (2008). Different turbulent dissipation models are also indicated (see text and Fig. 12).

to 25 pixels (see Fig. C.7). The obtained exponents $\tilde{\zeta}_p$ are plotted in Fig. 15. The error bars for SWO come from the uncertainty on the fit, while for PCA, the values and uncertainties are obtained by computing the structure functions from cubes reconstructed with $N = N_{\text{opt}} \pm 5$ components; the exponents and their uncertainties are respectively the mean and quadratic sum of the fit results and uncertainties for each value of N . The differences in $\tilde{\zeta}_p$ at $p = 5$ compared to the results of Hily-Blant et al. are 0.010 and 0.012 for PCA and SWO respectively. The three models follow the She & Lévy (1994) dissipation model.

9. Discussion

9.1. PCA and SWO as data reduction tools for large datasets

The use of PCA as a data compression tool is one of the main application in data science. The critical aspect in the present context is the convergence criteria which ensures that the meaningful signal is reproduced to a sufficient degree so as to enable the calculation of velocity centroid without the step of a window definition. The criteria we have built is based on the assumptions that the noise is Gaussian and homogeneous in the input data. In turn, this criteria proved very sensitive to failure of any of these assumptions. This may be useful to analyze quantitatively the noise behavior on large hyperspectral maps where the baseline subtraction is often the most critical step. For this step,

the method developed in the SWO approach provides a new, and robust, mean of automatic spectral window definition, including lines of sight with multiple velocity components, overlapping or well separated. We have successfully tested an automatic approach to spectral window definition on real data, by applying the SWO algorithm to spectra spatially averaged within small boxes to ensure a high SNR (Delcamp et al. 2024).

One of the benefit of the PCA method is that it only required two critical parameters: N_{opt} and an intensity threshold. For the first one, we developed a quantitative analysis allowing to retrieve it, while our benchmark on simulations gave a window of intensity threshold based on the noise of the original data cube. While important, this parameter does not impact as much the performances of SWO in retrieving the CV values. While there is not any critical parameter to be tuned for using this method, SWO is less robust than PCA at low SNR.

These methods were developed independently, without the intent of using both of them in the same time. However, the noise reduction produced by PCA can be use by SWO to produce more qualitative signal extraction. We highlight here that the present work does not follow this procedure : each method is used separately. The extraction of the turbulent properties of a molecular clouds face a lot of difficulties, from the projection of the velocity field on the plane of sky, to the small number of data available. Thus, it is of primordial importance to be careful in our estimations and to present error bars and uncertainties. The development of PCA and SWO begun with this difficulties in mind, with the will of bringing uncertainties where we don't have them by using them independently. Indeed, the PCA followed by the SWO method on the same data cube could lead to better results, but we will miss the opportunity to have a more reliable analysis.

9.2. Intermittency of molecular cloud turbulence

In the present work, two independent methods are developed to measure the centroid velocity and derive the structure functions and their scaling exponents. Their application to modeled clouds showed that the CVI distributions are very well recovered at large enough lags (see Fig. 9) but with differences between the measured and targeted distributions that increase towards the smallest lags.

The differences between the recovered and target structure functions at small lags are due to differences in the non-Gaussian wings of the CVI distributions (see Fig. 9), indicating that these large increments are associated to minute variations of the line profile that are improperly recovered by either PCA or SWO. This in turn indicates that the largest CVI, which are found at small lags, are most likely driven by spectral shape variations within the noise. This suggests that recovering these large increments is not only a matter of increasing the statistics, but also of improving the SNR. In addition, that the number of large increments is underestimated at small lags also suggests that the intermittency is stronger than anticipated, which would drive the exponents to lower values. This is beyond the scope of the present paper and will be explored in forthcoming studies based on larger numerical simulations of clouds.

9.3. ESS method and the internal coherence of structure functions

The ESS method of Benzi et al. (1993) has been used extensively in the context of terrestrial turbulence experiments, because it extends the range of scales over which a scaling of the velocity structure functions is observed, and thus improves the determination of the relative exponents $\tilde{\zeta}_p = \zeta_p/\zeta_3$. Yet, the theoretical foundations of the ESS method remain elusive (Arneodo et al. 1996; Meneveau 1996; Chakraborty et al. 2010). Still, this empirical method enables higher orders to be measured and thus provide an extended view of the dissipation range, with the price of not measuring ζ_3 . In the context of diffuse molecular clouds turbulence, the ESS was applied to this aim by Hily-Blant et al. (2008), giving access to orders $p > 2$ and thus allowing the comparison to intermittency models.

We have seen that the CVI distributions are better recovered at large lags than at the smallest ones. Yet, the velocity structure exponents $\tilde{\zeta}_p$ are always well recovered, including those of high orders which are the most sensitive to the small-scale statistics. This apparent contradiction is not due to projection effects since the comparison is done against the noise-free simulated cube and not on the genuine three-dimensional velocity field. Rather, this may reflect the use of the ESS method since the $\tilde{\zeta}_p$ are obtained through the fit of S_p versus S_3 . This may indicate that the ESS method corrects the lowest lags of the structure functions, probably because the third-order structure function presents the same biased values. In some sense, there appears to be a fortunate consistency in the measured velocity structure functions. Regardless, that the exponents $\tilde{\zeta}_p$ are correctly recovered is a strong support to use the ESS method to characterize the intermittency of molecular cloud turbulence. Yet, this gives no access to the value of ζ_3 which likely encapsulates characteristics of strongly compressible turbulence (Kritsuk et al. 2007, 2017).

9.4. Computation of high-order structure functions

High-order velocity structure functions are of utmost importance to characterize the properties of turbulence because they put the largest weight to the small-scale velocity gradients, those which carry the strongest signatures of intermittency and which are at the heart of turbulence dissipation. Yet, recovering these high- p $S_p(\ell)$ from experimental data is highly challenging and risky. The underlying difficulty, regardless of the use of the ESS method or not, is the convergence of the integral in Eq. (3). This is a long-standing debate and there is no theory providing a general criteria ensuring the computation of high-order structure functions. The study of Tennekes & Wyngaard (1972) summarized the problem at the time, leading to the conclusion that, their present state-of-the-art data-acquisition and processing techniques, would never allow $S_p(\ell)$ at orders $p > 4$ to be measured. Indeed, the higher the order, the larger the number of data needed to allow the integral to converge. Yet, few years later, the French school of turbulence obtained experimental data enabling the scaling of $S_p(\ell)$ up to order $p = 18$ or above to be measured (Anselmetti et al. 1984), while not closing the discussions however (Belin et al. 1996). Still recently, papers have tried to give general solutions to access the maximal order to compute (Dudok De Wit 2004). One of the problem is that the highest order accessible is not only a function of the number of data samples, but is thought to also depend on the properties of the turbulent field itself (DeMarco & Basu 2017).

In the present study, our analysis of the convergence of the velocity structure function follows Belin et al. (1996) and is

based on the behavior of the integrand in Eq. (3). As mentioned previously, the integrand has the typical shape shown in Figs. 10 and 14. At low orders, it reaches a clear maximum while, at high orders, the integrand becomes more scattered at large CVI increments, and eventually, at $p = 6$, it hardly reaches a maximum. In such conditions, the value of the integral is uncertain.

To see better the behavior of the integrand with increasing CVI (in absolute value), we compute its mean and dispersion in bins of 40 CVI. Figure 16 show the result for the simulated SNR=14 cube, where increasing and decreasing CVI are shown separately. This gives a close-up view of the evolution of the integrand. At low order, the integrand first increases and reaches a clear maximum, followed by a decrease down to zero, such that the integral converges. For order $p < 4$, the two parts do show the same convergence. Yet, at higher orders, the behavior of the integrand is not symmetrical between the positive and negative directions. Thus, for $p = 4$, the positive side converges (bottom) but the negative side (top) shows a large scatter of the integrand beyond the maximum. For $p = 5$ and 6, the integrand in the positive direction decreases to zero to within the uncertainties, while in the negative direction, the last point is not compatible with zero to within 2σ . We shall consider that convergence is obtained if at least one direction reaches zero to within 2σ . With this criteria, PCA and SWO do not lead to the same maximum order for the simulated cube with a SNR of 14, being $p_{\max} = 5$ with SWO and $p_{\max} = 6$ with PCA. In either case, the positive and negative directions have different maximum order.

Despite these differences on the velocity structure functions, there is no difference at $p = 6$ on the exponents between the PCA and SWO, on either the noise-free or SNR=14 cubes. Actually, it is at first sight unexpected that $p_{\max}=6$ is obtained with PCA for the SNR=14 cube, and $p_{\max}=5$ for the noise-free cube. In addition, it may seem surprising that the ζ_6/ζ_3 obtained with SWO or PCA are compatible, although only with PCA does the structure function converges. Furthermore, the values of $\tilde{\zeta}_p$ seem very consistent: thus, ζ_6/ζ_3 falls precisely on the dissipation model (here Boldyrev et al.) indicated by the $p \leq 5$ exponents (see Fig. 12). This suggests that our convergence criteria, based on the evolution of the integrand with increasing (decreasing) CVI, may be too restrictive. In addition, the use of the ESS method may also give more weight to correlations among the velocity structure functions of different orders, which could mitigate the convergence issues. This is however likely related to profound theoretical aspects of intermittency and is well beyond the scope of the present paper. Finally, convergence may not be achieved at a small lag (here $\ell = 2$) but well achieved at higher ℓ . This is shown in Fig. B.5 where the integrands, for $\ell = 7$ and $\ell = 25$ is seen to converge up to $p = 6$ in either positive or negative CVI. That the derived exponents show a high degree of consistency, even when marginal convergence is obtained at small lags, indicate that the convergence of a given order should be examined globally for a range of lags rather than requiring convergence at all lags separately. Future versions of `Unveil.jl` should improve the convergence criteria along these lines. In the current version, it is up to the user, on the basis of plots such as Fig. 16, to decide the value of $p_{\max}$ that makes physical sense.

9.5. Uncertainties on the statistical analysis

Estimating the uncertainties of an analysis method is always difficult. One important aspect of the `Unveil.jl` pipeline is to provide two, fully independent methods to compute centroid velocity. This is at the heart of the proposed approach to estimate the

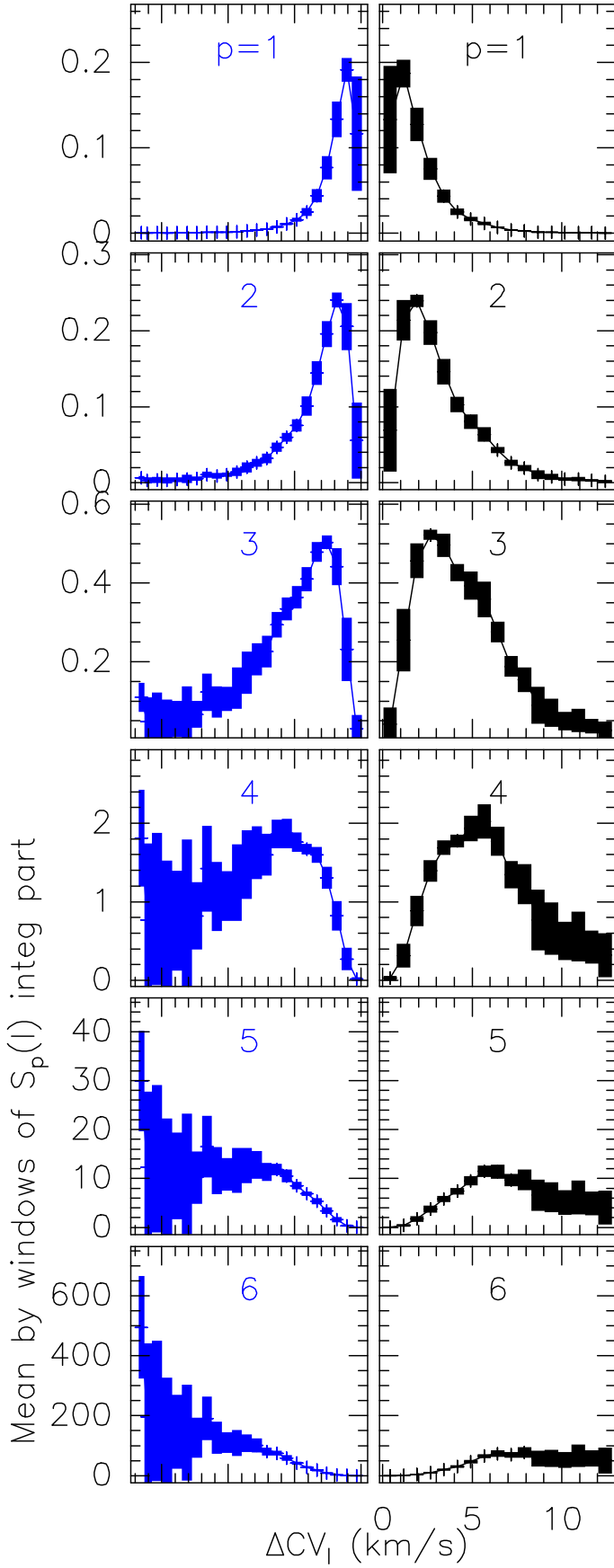


Fig. 16. The integrand in Eq. (3) as a function of the CVI, for the simulated SNR=14 cube at lag $\ell = 2$. The integrand was binned by steps of 40 values; shown are the mean as scatter points and dispersion as errorbar of each bin. The integrand is computed separately for decreasing (top) and decreasing (bottom) CVI. **the figure changed**

uncertainties on the derived statistical analysis, such as the CVI distribution, or velocity structure function exponents $\tilde{\zeta}_p$.

Regarding to the uncertainty on the number of PCs to use, the CV and further computations should be done on a data cube reconstructed with different numbers of N . In this sense, PCA brings an easily reachable estimation of the uncertainty of these computations.

In addition, the PCA method provides an internal consistency check because the number of components used to reconstruct the data cube is a free parameter: our convergence criteria Eq. (11) provides the user with a robust estimate of the optimal value regarding the properties of the reconstructed noise. The adopted criteria is not supposed to represent the only way to reach PCA convergence. However, it is robust, as was presented with our benchmark on simulated data cubes. Based on the value of N_{opt} , it is possible to vary the number of components around N_{opt} , and explore the impact on the statistical properties such as the $\tilde{\zeta}_p$. This in turn enable the uncertainties to be quantitatively estimated.

9.6. Observational requirements

The present work may provide guidelines for future observations of turbulence in molecular clouds, especially in terms of the aimed signal-to-noise ratio. In turn, the SNR study presented above indicates that PCA may be better suited to a given data set depending on the SNR.

For observational data with a SNR of 4, PCA should be used before SWO, but the choice of N_{opt} would be difficult to made. However, the independence of the two methods encourages to use them together, thus choosing a SNR ≥ 8 . With a SNR ≥ 12 , our estimations suggest that PCA and SWO will conduct to less than 1% of CV values different from noise-free observations of $1\delta v$. An intensity threshold should be used, with a value between 2 and 3 times the noise of the cube.

While always recommended at least for easier data reduction or more scientifically robust region-by-region comparison, these two methods do not require a homogeneous spatial noise distribution. With its spectra-per-spectra treatment, each spectrum in the SWO procedure is independent of the others. An option allows SWO to process smoothed spectra to enhance the cube SNR, but this is not a mandatory adjustment that would affect its efficacy. PCA is not disadvantaged in a data cube with inhomogeneous noise; however, this feature will become apparent during convergence analysis. Easily identified in the data projected maps and the metric (see Fig. C.2 and C.1), the choice of N_{opt} is not affected. The limiting case where the noise variance in specific regions of the data cube with a low SNR could be correlated with the variance of the signal elsewhere in the cube is unlikely and undesirable for any scientific analysis.

We might assume that PCA would perform better when working on similar spectra separated into multiple cubes rather than a single global cube. To verify this, we used PCA to sort spectra by phenotype, leveraging its ability to sort variance by linear combination. Three cases were tested: a cube divided into four parts without any assumptions about their boundaries, the cube divided into four where spectra shapes were similar within each, and the global cube without treatment. With $N \ll N_{\text{opt}}$, we did not notice any significant performance differences. Each case presented spectra with different morphologies compared to the original, and no method consistently outperformed the others. When N_{opt} was reached, all methods accurately reproduced the spectra shape and the same noise level. We concluded that

PCA does not perform better if spectra are similar within a given data cube, thus no pre-treatment data analysis on the cube is required.

10. Conclusions

The present work has presented the `Unveil.jl` pipeline, which offers new, and original tools to explore the statistical properties of turbulence in molecular clouds. Because the cornerstone of such analysis is the measurement of an estimator of the projected velocity, the heart of `Unveil.jl` is the calculation of the centroid velocity. By providing two, fully independent methods, `Unveil.jl` enables the uncertainties to be measured, and thus provide robust CV maps.

The two CV measurement methods at the heart of `Unveil.jl` are new and original. One relies on the use of the *Principal Components Analysis* (PCA) to reduce the dimensionality of the hyperspectral datacube and thus artificially improve its SNR. The main strength of the PCA is the decomposition on an orthogonal set of eigenvectors, thus providing a non-degenerate decomposition, unlike other methods using e.g. a Gaussian decomposition of spectral lines. The gain of RMS carried by this method can reach a factor 30. The PCA allows the centroid velocity to be computed with no need to define a spectral window for each spectrum, although an intensity threshold is applied. On the other hand, the *Spectral Window Optimization* (SWO) method proposes an improved approach to automatic spectral windowing, that is robust against multiple velocity components spectra. The amount of **correct** CV is larger than 99% and 96% for PCA and SWO respectively, on a cube with a SNR of 14. In turn, the present study indicates that a SNR of 12 is needed to obtain converged measurements for both PCA and SWO, of the CV, CVI, and structure functions.

It was found that the PCA is intrinsically more sensitive than SWO to the spatial homogeneity of the noise in the input data. Actually, the SWO method can also be used to perform automatic baseline subtraction of heterodyne hyperspectral cubes of molecular clouds, and thus to further improve the performance of the PCA method (Delcamp et al. 2024).

The pipeline also provides tools to compute the increments of CV over a range of lags, and to build their probability density functions. This product is also the starting point for the calculation of the velocity structure functions of various orders. A significant effort was devoted to the analysis of the convergence of the calculation of these structure functions, especially of high orders since this is a well-known challenge in experimental turbulence. We thus built new methods to inspect the convergence of the structure functions. The measurement of their scaling exponents was done exclusively with the ESS method, thus providing only relative exponents $\tilde{\zeta}_p = \zeta_p/\zeta_3$, although the pipeline does not restrict the fitting of the $S_p(\ell)$ to the ESS method.

The pipeline was finally tested on the Polaris Molecular Cloud for which the structure function analysis was performed with the pipeline of (Pety & Falgarone 2003). The results of Hily-Blant et al. (2008) are recovered, although the convergence of the $p = 6$ structure function is questioned by the present analysis. Nevertheless, the agreement with the mildly compressible scaling of structure function exponents (She & Lévéque 1994; Porter et al. 2002) is recovered.

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Appendix A: Convergence of the PCA

The fundamental assumption in the adopted convergence criteria (cf Eqs. (10) and (11)) is that the intensity in eigenimages with $N > N_{\text{opt}}$ behaves as a Gaussian noise. We have validated this assumption by applying the PCA to three simulated data cubes: one with only noise (no signal), one with signal only (no noise), and one with signal plus noise. We also examine the application of the PCA to the Polaris data cube. The results are presented Fig. A.1 where we show the probability density functions of the intensity in three eigenchannels, $N = 1, 20$, and 50 .

The intensity in the pure-noise cube (top row) follows a Gaussian distribution in all three eigen channels. In contrast, the distributions always show non-gaussian characteristics in the signal only case; for $N = 1$, the majority of the variance is negative. For the noise-free cube (second row), the distribution is never Gaussian. The noise plus signal cube (third row) presents non-Gaussian wings at $N = 1$, as expected, but is Gaussian for $N \geq 20$. In the case of the Polaris data cube (bottom row), the distribution in the first eigenimage ($N = 1$) is clearly non-Gaussian, but at $N = 20$, the distribution is already very close to a Gaussian. At $N = 50$, the deviations from a Gaussian distribution can no longer be examined by eyes and require the use of moments, which confirm its Gaussian nature. With this example, we see that the eigenimages associated with eigenvectors reproducing signals will present non-Gaussian distributions, while, beyond a certain range, the intensity in eigenimages are very close to Gaussian distributions.

Figure A.2 exemplifies the spatial behavior of the intensity in the eigenimages for the simulated cube signal+noise with a SNR of 14. For this cube, the value of N_{opt} is 19. The first eigenimage shows the average intensity in the map, since each pixel tells how much its spectrum resembles the spatially-averaged spectrum. The subsequent eigenimages are a spatial representation of the variations around the mean spectrum. We thus see at $N = 2$ that the top half-part present mainly positive differences with the $N = 1$, and negative for the bottom half-part. The eigenimages show clear structures for $N \leq N_{\text{opt}}$, which are more prominent in both amplitude and size as N decreases. At $N > N_{\text{opt}}$, the intensity map becomes featureless and can be interpreted as noise.

Appendix B: Simulated data cubes

Appendix B.1: Simulated spectra

The simulated data cubes used in the benchmark of `Unveil.jl` contain a wide variety of spectra, as may be seen in Fig. B.1. The simplest spectra have a Gaussian-like shape, which can be seen on the bottom-right part. Others have faint wings, seen on the bottom-left part. Multiple distincts regions presents multiple velocity components, clearly distinguishable for some, while others are mixed.

Appendix B.2: Linewidth-size relation from CVI distributions

We present in Fig. B.2 the dispersion of the CVI distributions computed at several lags, for the simulated cubes with SNR=4, 8, 12, and 14. The CVI distributions for the SNR=14 and the noise free cube at 5 lags where presented Fig. 9. They are compared to the CVI computed for the corresponding noise-free cube.

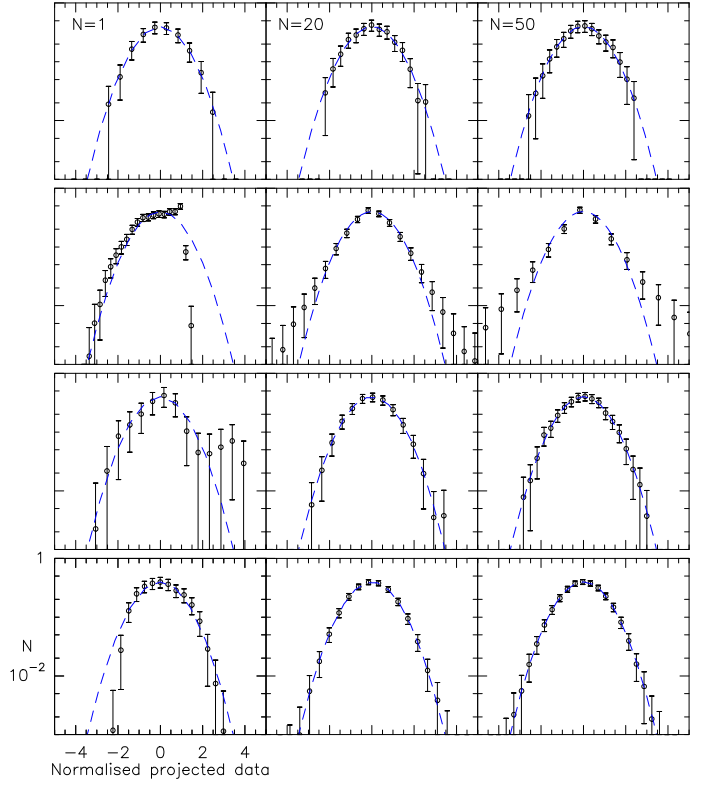


Fig. A.1. Distribution of the intensity in three eigenimages ($N = 1, 20$, and 50 , from left to right) from the PCA applied to four cubes. From top to bottom: pure-noise, pure-signal, signal plus noise, and Polaris (cf Section 8). For the simulated cube, $N_{\text{opt}}=19$, and $N_{\text{opt}}=29$ for Polaris. The distribution are centered and normalised to unit dispersion. Centered, normalized Gaussians are plotted in dash blue lines. Uncertainties on the bin numbers are statistical.

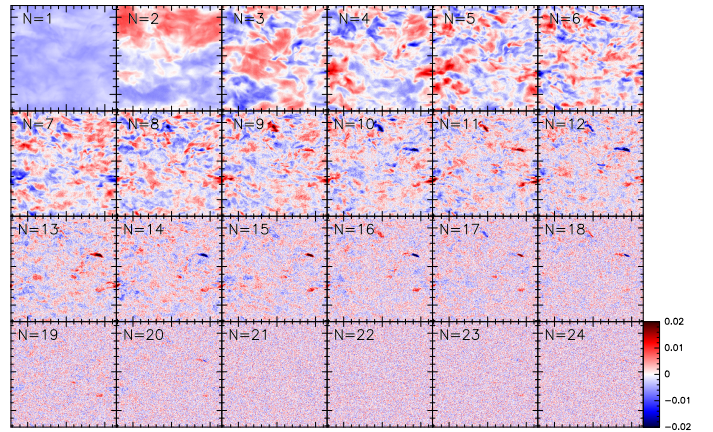


Fig. A.2. Simulated cube at SNR=14 projected on the 24 first eigenimages. The color scale is identical for each subplot.

Appendix B.3: Structure functions fits

The structure functions of the simulated data cube are plotted in ESS in Fig. B.3, following the same caption as Fig. 11. A power-law behavior is found in all cases, except for SWO for the cubes with a SNR of 4, and also 8 at high order. The ESS method allows to increase the range over which a power law can be fitted. Indeed, while only the first few data points could be used without the ESS assumption (see Fig. 11), here almost all the points can be used.

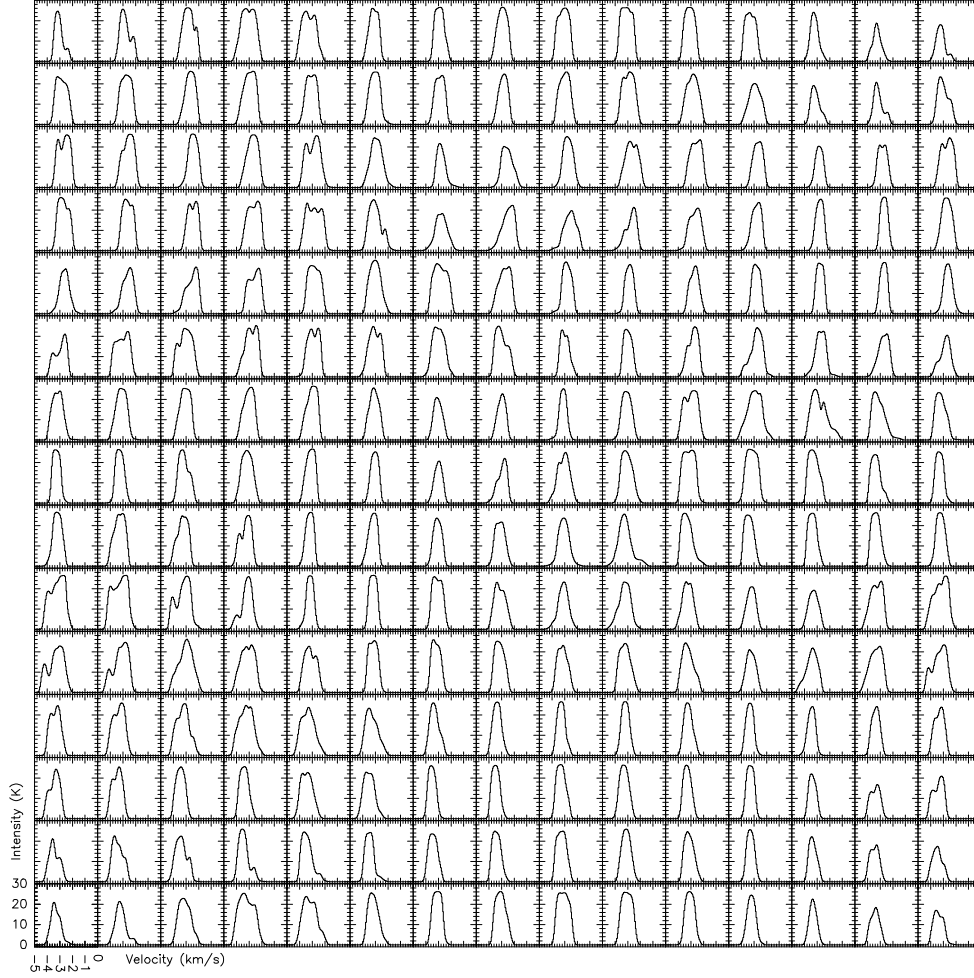


Fig. B.1. The simulated CO(1-0) lines profiles, based on the [Cho & Lazarian \(2003\)](#); [Burkhart et al. \(2020\)](#) simulation. The displayed spectra have been averaged by boxes of 18×18 . The velocity is in km s^{-1} , and the intensity in K. See also Fig. 1.

Appendix B.4: ESS exponents

We show in Fig. B.4 the difference between the ESS exponents $\tilde{\zeta}_p$ from the noise-free cube and those measured with the SWO and PCA methods on cubes with SNR of 4 to 14. The corresponding exponents values are shown in Fig. 12. The vertical scale is the same for SNR=8, 12, and 14. It is clear that the values from the PCA are much less sensitive to the SNR than from SWO, and that both methods do equally well for data with SNR larger than 8.

Appendix B.5: Structure function convergence

The convergence of the structure functions is examined at various lags using the SNR=14 simulated cube in Fig. B.5.

Appendix C: Analysis of the Polaris field

Appendix C.1: PCA and convergence

The result of the PCA applied to the Polaris data is shown in Fig. C.1. The same evolution as described for A.2 can be seen here. In particular, the first channel is very similar to the integrated intensity map of [Hily-Blant et al. \(2008\)](#). Specific structures patterns easily observed in the PPV cube are recovered in $N \leq 10$. Regions higher than $\Delta l_{ii} \approx -5'$ present distinct evolution than the rest of the map at $N \geq 18$. This is also a region

with a higher noise than in the rest of the map (see Fig. C.3), and is thus described by different proportions of variance than the others regions. The noisy character of the intensity in channel maps beyond ≈ 30 is clearly apparent, and is at the root of the convergence criteria. The metric is shown in Fig. C.2. There is a rapid fall of the metric from 1 to 20 components, and a plateau until $N \approx 150$. The metric then rises beyond $N \approx 170$: this non-generic behavior is due to spatial inhomogeneities of the noise in the input ppv cube (see Fig. C.3).

Appendix C.2: Centroid velocity map

The CV maps computed with PCA and SWO, and from [Hily-Blant et al. \(2008\)](#), are presented in Fig. C.4. PCA and SWO allow to recover all the structures already seen with the earlier method. A quantitative analysis, with a pixel-per-pixel differences between the [Hily-Blant et al. \(2008\)](#) CV map and the CV maps of each method, is summarized in Fig. C.5 which shows the histogram of the differences. We see that the largest differences with PCA (top left panel) come from the eastern region, at low Δl_{ii} , in a region with the lowest SNR (see Fig. C.3). Negative differences in PCA seem to follow the South-East to North-West elongated structure, thus PCA overestimate the CV values in this region compared to the previous method. For SWO, the differences are smaller but appear more structured spatially. The histograms of the differences (bottom panel) show that the differ-

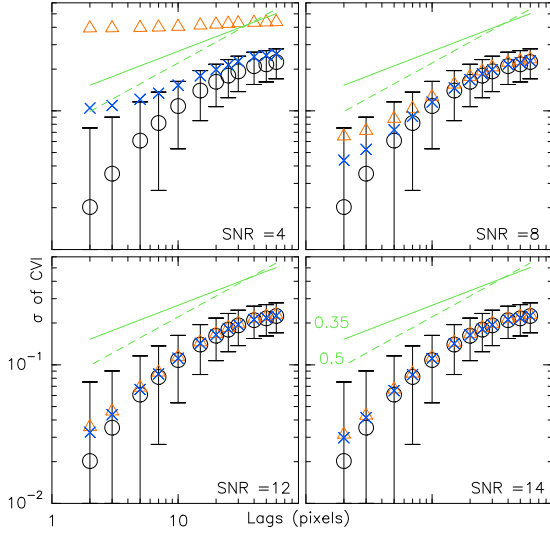


Fig. B.2. Dispersion of the CVI distributions at multiple lags ℓ and for each of the SNR cubes, treated by SWO (orange triangles), reconstructed by PCA (blue crosses), and from the noise-free cube (circles). Uncertainties are $\pm 1\delta v$ around the noise-free values. Two power-law are given in green : one at 0.35, the other at 0.5 (following She & Lévéque (1994)).

ences are larger from the PCA method than SWO, and that they are also more numerous. This is most likely due to the spatial inhomogeneity of the noise to which the PCA is more sensitive; indeed, while SWO treats each spectrum independently, that is not the case with the PCA method which first computes the eigenvectors using all spectra together through the total amount of variance reproduced. However, no matter which method is compared to the one from Hily-Blant et al. (2008), 99% of these differences lies within this uncertainty

Appendix C.3: CVI and velocity structure functions

A sanity check of the CVI and velocity structure function calculation in `Unveil.jl` was performed, which consisted in computing these quantities from the CV map of (Hily-Blant et al. 2008). The thus obtained $S_p(\ell)$ were fitted, for $\ell = 3$ to 25 pixels, and the ESS exponents retrieved. We obtained excellent agreement between the results from the CVI and $S_p(\ell)$ processing in `Unveil.jl`, and the earlier values of Hily-Blant et al. (2008) (see Fig. C.6). The exponents are also compared to theoretical predictions for homogeneous turbulence (K41), incompressible or mildly compressible turbulence (She & Lévéque 1994; Porter et al. 2002), and shock dominated magnetized turbulence (Boldyrev et al. 2002). The three calculations confirm the solenoid-dominated turbulence in the Polaris Flare; in particular, the differences between the values of ζ_p at $p = 6$ are negligible in comparison to those separating each dissipation model. We can conclude that the CVI and $S_p(\ell)$ computations of `Unveil.jl` are identical to the ones of Hily-Blant et al. (2008).

The first 6 orders of the structures functions computed on the Polaris data cube, obtained with PCA and SWO, are plotted with the ESS method Fig. C.7. The range for the fit cover all the lags probed, giving one slope with low uncertainty on the fit.

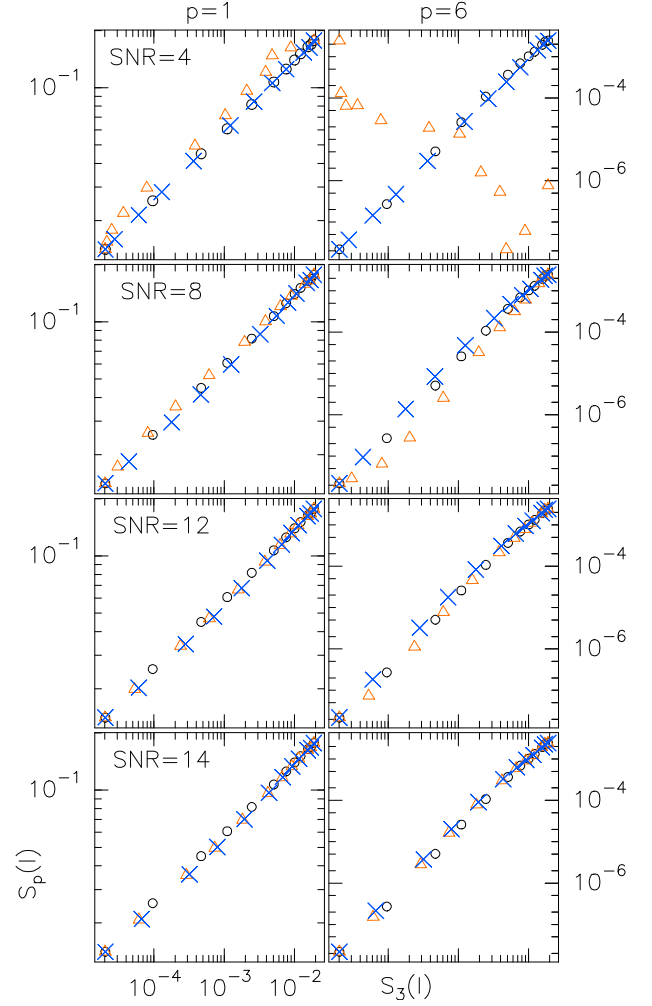


Fig. B.3. Structure functions plotted against $S_3(\ell)$ of the simulated cubes, treated and reconstructed by SWO and PCA with N_{opt} in orange triangles and blue crosses respectively, and the noise-free cube as black circles. Axis limits are adapted to each data. The orders are $p = 1$ on the first column, and $p = 6$ on the second column.

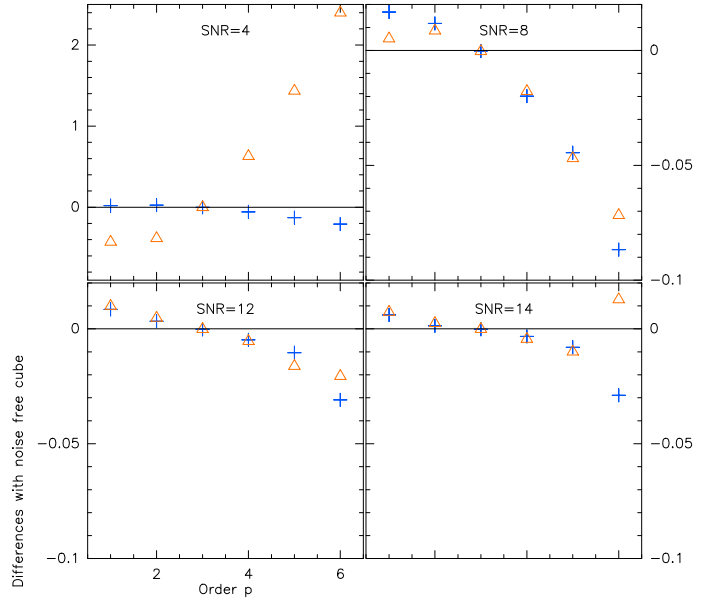


Fig. B.4. Differences of the exponents from PCA and SWO with exponents of the noise-free cube (see also Fig. 12). A horizontal line is drawn at 0 for reference.

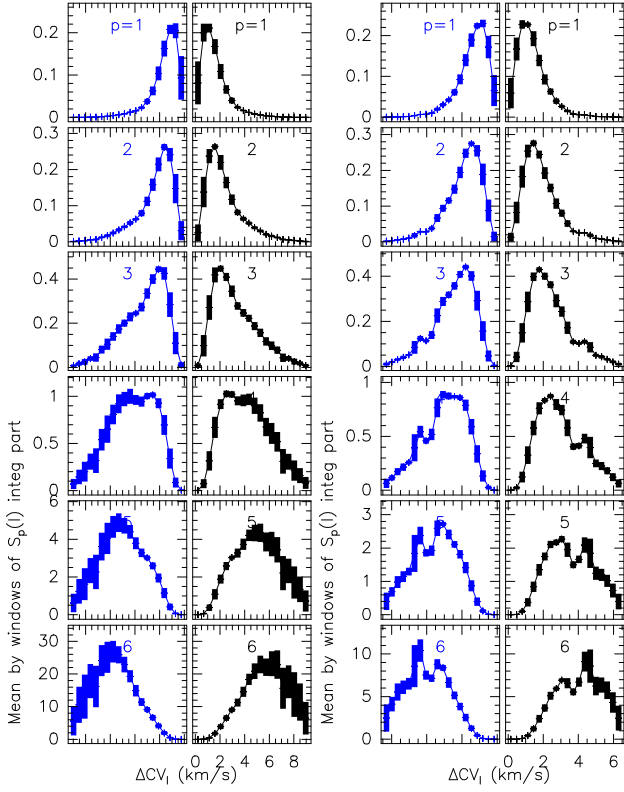


Fig. B.5. Integrand as a function of the CVI, for the SNR=14 simulated cube, for a lag $\ell = 7$ pixels (**left two columns**) and $\ell = 25$ pixels (**right two columns**). **figure changed**

Appendix D: Technical details on Unveil.jl

Appendix D.1: CVI and $S_p(\ell)$ implementation

For statistical analysis, the relative difference is used, but for graphical extraction of velocity shears, we will use their absolute values. The CVI are computed by rotations of the CV map. The number of rotations necessary T depends of the lag l following : $T = 2\pi / \arctan(1/l)$. Others methods are in development to check the possible bias imposed by the sample of a rotating map. Preliminary results show minor differences, lower than the velocity resolution. However, this analysis require further investigations.

Two equations giving same results are computed in Unveil.jl, with $p(\Delta V_l)$ the probability distribution function of all the CVI rotations:

$$S_p(l) = \int_{\Delta V_l} p(\Delta V_l) \Delta V_l^p d\Delta V_l = \langle |\Delta V_l|^p \rangle \quad (\text{D.1})$$

While the second one is faster to compute, the first one will be used to verify and check convergence of the process.

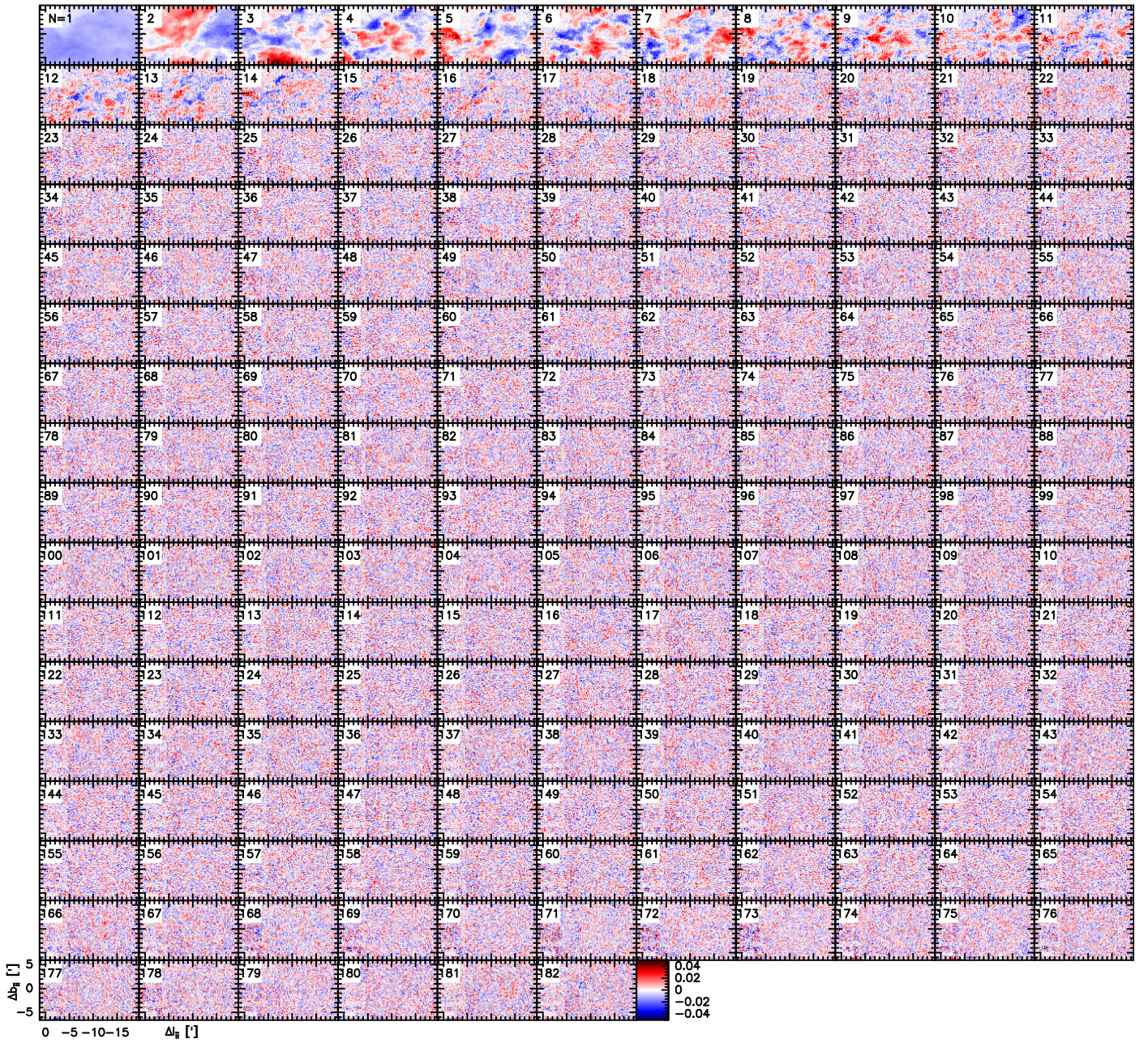


Fig. C.1. Eigenimages of the Polaris observational data from $N = 1$ to $N = 182$. The color scale is identical for each subplot.

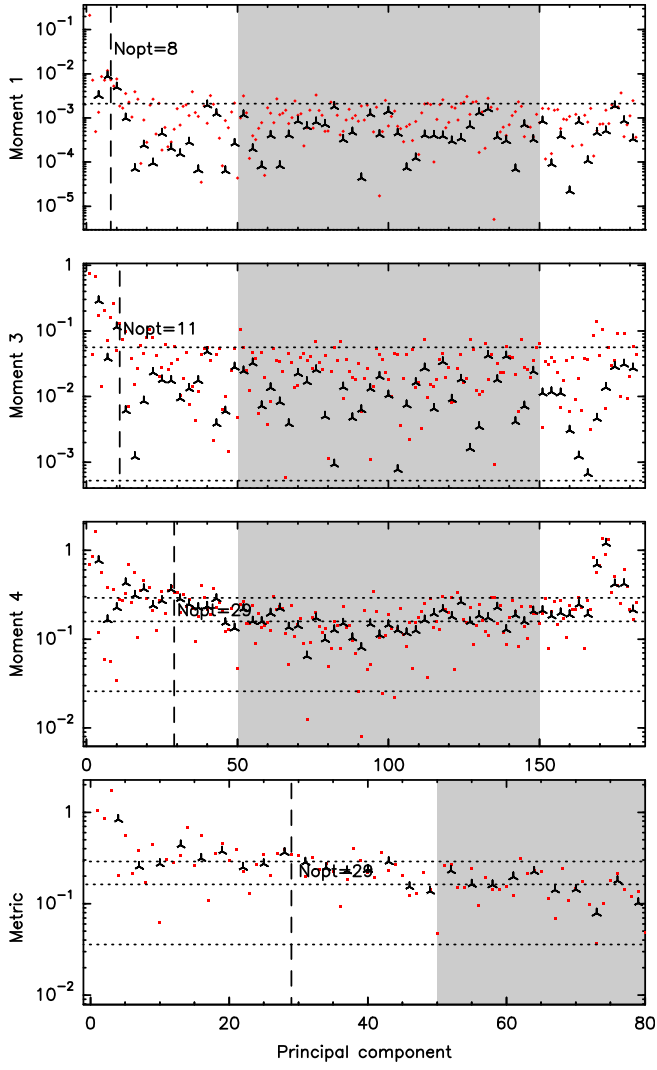


Fig. C.2. Same as Fig. 4 for the Polaris field. The value of N_{opt} is 29.

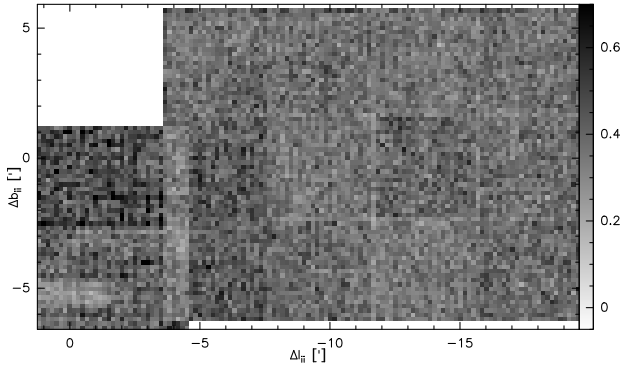


Fig. C.3. Spatial distribution of the intensity noise (in K) of the Polaris data cube. The spatial offsets (in arcminutes) are in galactic coordinates, with the same origin as in [Hily-Blant et al. \(2008\)](#).

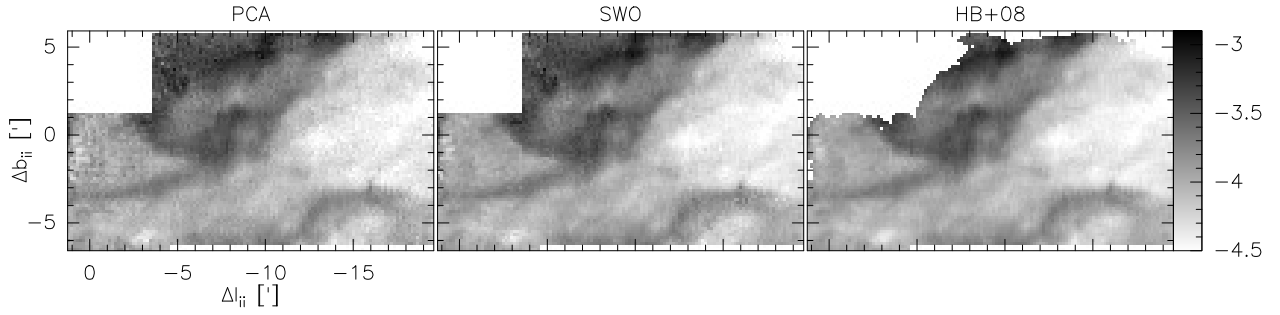


Fig. C.4. CV maps, computed from the PCA reconstructed cube with $N_{\text{opt}}=29$, from SWO, and obtained in Hily-Blant et al. (2008). The color scale is identical for each panel and in units of km s^{-1} . For the PCA and SWO processes, we kept all spectra without removing those with low SNR like in Hily-Blant et al. (2008).

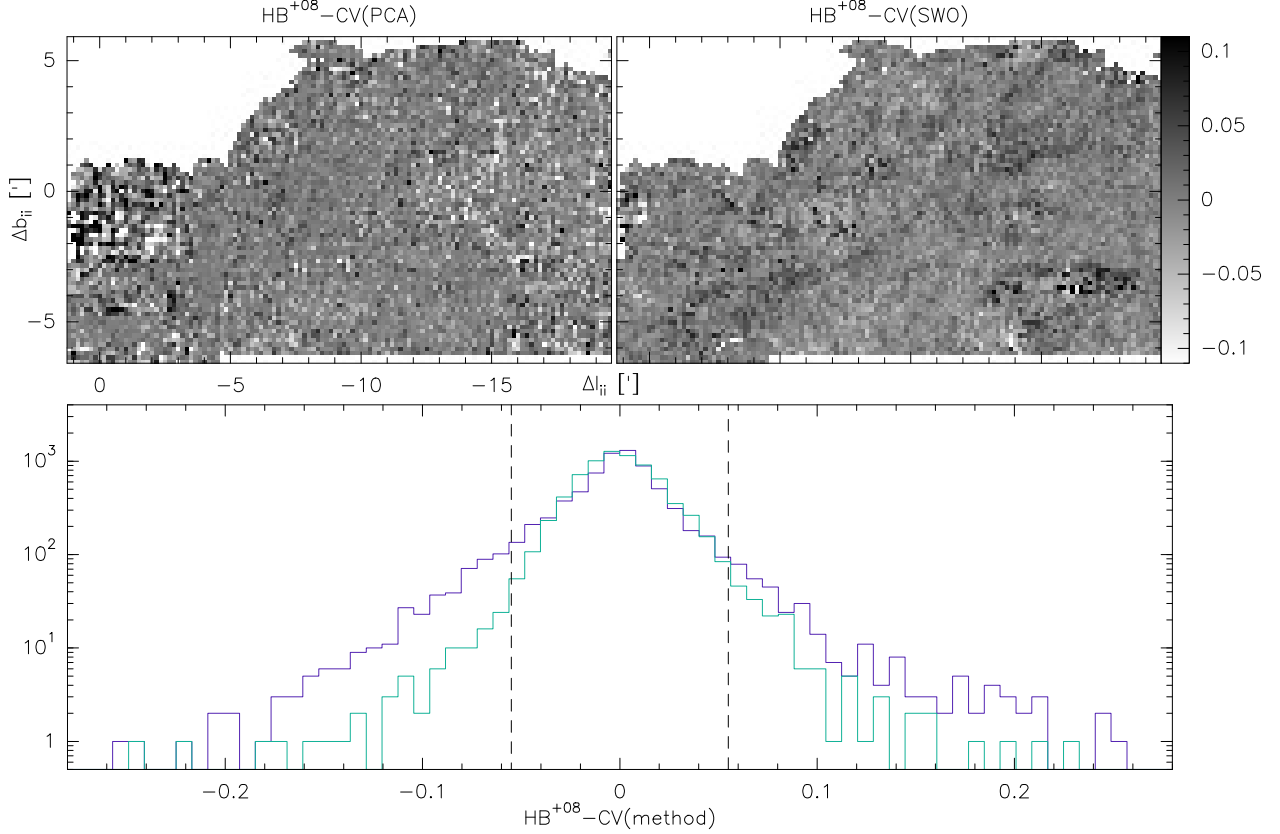


Fig. C.5. Differences of the CV map (in km s^{-1}) between Hily-Blant et al. (2008) and the two methods (PCA with $N_{\text{opt}}=29$ and SWO at left and right respectively). The color scale is identical for the two panels, with extremas being equal to 2 times the velocity resolution (e.g. 0.055 km s^{-1}). The third panel is the histogram of the differences for the two methods. The black dashed lines are the velocity resolution.

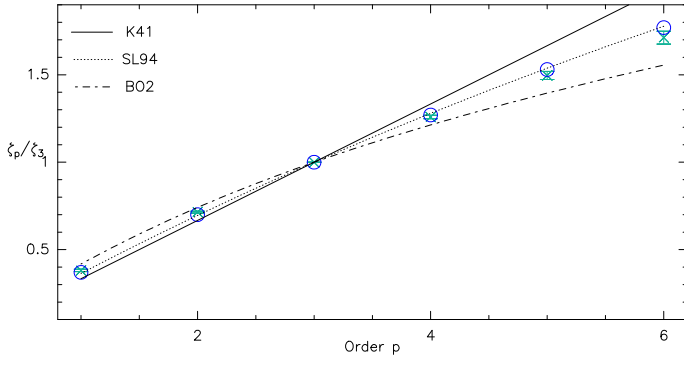


Fig. C.6. Exponents $\tilde{\xi}_p$ of the velocity structure functions computed on the Polaris data, and using the CV map of Hily-Blant et al. (2008). In blue circles are the values from Hily-Blant et al. (2008), in green crosses are the exponents obtained after the CVI and $S_p(\ell)$ processes of Unveil.jl from the Hily-Blant et al. (2008) CV map. The error bars correspond to the uncertainty on the fit.

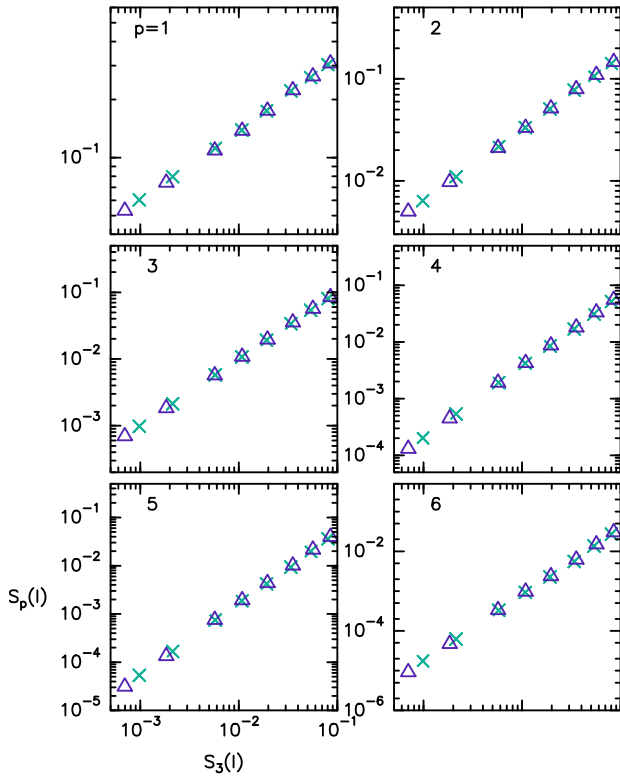


Fig. C.7. Structures functions plotted against $S_3(l)$ for Polaris data treated and reconstructed by SWO and PCA with N_{opt} in purple triangles and green crosses respectively, and the noise-free cube as black circles. Axis limits are different for each data.